

From soluble iodine to functional blockage in zinc–iodine flow-battery positive electrodes

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The positive iodine electrode limits the practical current density and accessible capacity of zinc–iodine flow batteries (ZIFBs), yet the first appearance of oxidized iodine is not the failure event. We develop a multiscale positive-electrode mechanism that follows iodine from soluble oxidized species to retained native $I_2(s)$ and, finally, to *functional blockage* through a morphology-controlled accessibility loss. The framework separates three states that are usually conflated: iodine *present* in solution, iodine *retained* at the carbon felt, and iodine that *blocks* active reaction and transport pathways. A native continuum scaffold supplies the baseline positive-electrode iodine-state clock, summarized by a within-cycle surface stress $S_{\text{base}} = (\Pi_{\text{inv}} + \Pi_{\text{gen}})/\beta_{\text{intra}}$, with Π_{gen} the generation/transport load, Π_{inv} the oxidized-iodine inventory and β_{intra} the running speciation buffer. Raw full-cell composition axes show that β_{intra} alone cannot carry electrolyte composition: supporting bromide raises the capacity ceiling $\sim 2.5\times$, whereas added ZnI_2 lowers capacity monotonically. We therefore extend the baseline stress to $S_{\text{eff}} = S_{\text{base}} \chi_{ZnI_2}^{\text{rel}}/\Lambda_{\text{Br}}^{\text{rel}}$, where $\Lambda_{\text{Br}}^{\text{rel}}$ is a baseline-normalized *apparent iodine-accommodation* factor and $\chi_{ZnI_2}^{\text{rel}}$ a weakly identified excess-iodide stress index, both calibrated separately from the continuum scaffold and not microscopic speciation constants. Passivation is governed by *two thresholds* — precipitation onset and functional blocking — whose finite separation ($\sim 17 \text{ mAh cm}^{-2}$ at the half-blocking criterion) makes “*retained iodine is not blocking iodine*” quantitative, and an independently measured iodine-dissolution rate sets the current scale. The design rule is to lower the generation stress, preserve within-cycle buffering, and raise iodine accommodation rather than simply increase total iodine inventory.

zinc–iodine flow battery, iodine passivation, super-saturation, porous electrode, dimensionless model

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1 Introduction

Zinc–iodine flow batteries (ZIFBs) couple a soluble, abundant and high-potential iodide/triiodide catholyte to a zinc anode, making them attractive for low-cost stationary storage [1–6], with recent work pushing toward practical, high-areal-loading cells [7]. In practice, however, the positive iodine electrode caps both the sustainable current density (typically $< 100 \text{ mA cm}^{-2}$) and the accessible areal capacity [5, 8], and this is not simply a question of how much iodine is present. During charge the catholyte can darken with oxidized iodine long before the cell reaches voltage runaway—the I_2 gener-

ated on charge is taken up by complexation and dissolution, so its mere appearance is plainly not the failure event. A useful mechanism must therefore distinguish iodine that is *present* in solution, iodine *retained* in the positive felt, and iodine that has become *functionally blocking*; a consistent lesson from the most effective recent electrolytes is that controlling the catholyte’s halide speciation—how iodine is complexed, buffered and solvated—matters more than the iodine amount [9–11].

The molecular origin of this distinction is the poor aqueous solubility of neutral I_2 . Charging generates I_2 from iodide, $2I^- \rightarrow I_2 + 2e^-$, after which the neutral intermediate

is buffered by I^-/I_3^- complexation ($K \approx 698 \text{ M}^{-1}$ [4, 12]) and, in bromide-bearing electrolytes, by I_2Br^- and related bromide accommodation [3, 8]. If dissolution and removal keep pace with generation, oxidized iodine stays soluble or reversible; if the free- I_2 stress exceeds the local accommodation capacity, native $I_2(s)$ forms in the porous carbon, as *operando* EQCM on nanoporous carbon directly resolves [13, 14]. Even then, retained $I_2(s)$ is not automatically passivating: it must overlap active reaction, transport or flow pathways before it exacts a voltage and capacity penalty; a passivating iodine film that raises the positive-electrode charge-transfer resistance, adding hundreds of millivolts of overpotential at high current density, has indeed been resolved directly [15]. Zhao *et al.* [16] quantified the underlying dissolution limitation in a related ZnI_2 flow cell, measuring a heterogeneous rate constant $k \sim 10^{-6} \text{ mol cm}^{-2} \text{ s}^{-1}$ and confirming an oscillating, dissolution-limited current.

Two experimental observations motivate the separation. First, compression is a *structural* paradox (Fig. 1a): the most-compressed felt reaches voltage runaway at the *lowest* usable capacity even though it is predicted to accumulate the *most* local solid iodine, so iodine inventory and blocking strength need not scale together. Second, a raw-data audit of composition-controlled full cells shows that electrolyte chemistry cannot be carried by the traditional speciation buffer β alone: at fixed 1 M ZnI_2 , raising NH_4Br from 0 to 2–4 M lifts the capacity ceiling $\sim 2.5\times$ and then saturates, whereas at fixed 4 M NH_4Br , raising ZnI_2 from 1 to 3 M lowers capacity $\sim 2.2\times$. The β factor, dominated by I_2/I^- complexation, reproduces neither the magnitude nor the direction of these cross-composition trends; bromide must instead enter the capacity-level model through a separate *apparent iodine-accommodation* term, while β remains the *within-cycle* speciation buffer.

Here we build a positive-electrode mechanism framework around these facts. A native continuum scaffold supplies the positive-electrode iodine-state fields under full-cell boundary conditions; from them a reduced stress coordinate $S_{\text{base}} = (\Pi_{\text{inv}} + \Pi_{\text{gen}})/\beta_{\text{intra}}$ extracts the within-cycle competition between oxidized-iodine inventory, generation/transport load and the running speciation buffer. A composition extension $S_{\text{eff}} = S_{\text{base}} \chi_{ZnI_2}^{\text{rel}}/\Lambda_{Br}^{\text{rel}}$ introduces the experimentally required bromide-accommodation and excess-iodide terms without treating them as microscopic speciation constants, and morphology closures map the native $I_2(s)$ inventory to an effective accessibility loss θ_{eff} , so that precipitation onset and functional blocking become separate thresholds. Because a cell-average voltage averages over strong local gradients, resolving this state requires the continuum scaffold rather than the terminal voltage alone [17, 18].

The rest of the model stack supplies the physical bounds and observables: density-functional theory identifies oxygenated carbon motifs as iodine-retention priors, molecular dynamics bounds the diffusivity of oxidized-iodine carriers, and fibre and pore-network models translate retained solid into accessibility and permeability consequences, while a reduced voltage layer tests whether the rising charge curve follows the positive-electrode state clock. We deliberately bound the scope: in single-cell zinc-iodine (hybrid) flow batteries the negative-electrode/membrane channel—zinc compaction and dendrite-driven separator perforation—is frequently the lifetime-limiting failure and is largely *composition-independent*, the excess cell voltage there tracking the membrane rather than the electrodes [19]; this is mechanistically distinct from the *composition-organized*, charge-transfer/accessibility-limited positive-electrode channel modelled here. Inter-electrode crossover and shuttle, addressed by membrane and separator design [20, 21], form a further complementary loss channel; the final noisy failure, discrete pore-throat clogging and pump-pressure runaway are treated as post-onset or out-of-scope. The paper thus does not claim a complete production-cell model: it tests a specific positive-electrode mechanism—soluble iodine becomes harmful only when retained iodine crosses a morphology-dependent boundary into functional blockage—and distils the model stack into design rules.

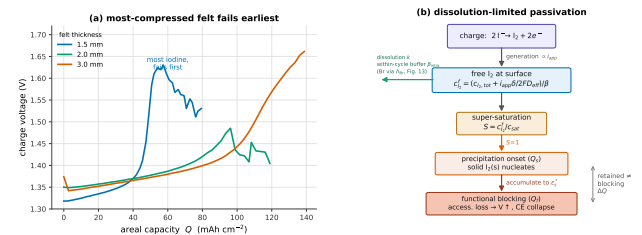


Figure 1 (Color online) **Iodine amount is not the penalty; functional blockage is a state transition.** (a) Charge curves from a compression series at fixed current density: the most-compressed felt reaches voltage runaway at the lowest usable capacity, even though the model predicts more local retained iodine. This motivates, but does not by itself prove, the separation between retained iodine and blocking iodine. (b) Mechanism formalized in Sec. 2. Charge generates oxidized iodine; within a given electrolyte the within-cycle buffer β_{intra} buffers the free- I_2 monomer and sets the baseline surface stress S_{base} . When S_{base} reaches unity, native $I_2(s)$ is retained (precipitation onset Q_s); functional blocking (Q_f) occurs only after the retained solid crosses the accessibility threshold θ_s^* , and the gap ΔQ is retained-but-not-blocking iodine. Cross-composition bromide effects are carried separately by the iodine-accommodation factor $\Lambda_{Br}^{\text{rel}}$ (Fig. 13).

2 Model framework and reduced iodine-state theory

Passivation is not governed by retained iodine alone, but by whether the *free* iodine exceeds its buffered solubility and whether the resulting solid then crosses an accessibility threshold. Our aim is to compress this picture into a single closed-form surface stress that the continuum solution obeys up to the stated normalization, and that predicts when failure occurs. We set out the physical picture and assumptions (§2.1), state the governing balances the continuum scaffold actually solves (§2.2), reduce the reaction front to a super-saturation stress (§2.3), define the two failure thresholds (§2.5), fix the dissolution-limited magnitudes (§2.6), and record where the closures come from (§2.8). Symbols are collected in Table 1.

Table 1 Principal symbols and baseline values ($c_{I^-,0} = 1688 \text{ mol m}^{-3}$, $i_{\text{app}} = 400 \text{ A m}^{-2}$, $T = 298 \text{ K}$).

symbol	meaning	value / units
n, F	electrons per I_2 ; Faraday	2; 96 485 C mol ⁻¹
$c_{I_2,\text{tot}}, c_{I_2}^f$	total / free dissolved I_2	mol m ⁻³
c_{sat}	I_2 saturation reference	1.33 mol m ⁻³
β	complexation buffer $1 + K_I c_{I^-} + K_{Br} c_{Br^-}$	554–1263
D_0, D_{eff}	free / Bruggeman I_2 diffusivity	0.5×10^{-9} ; $D_0 \varepsilon^{1.5} \text{ m}^2 \text{ s}^{-1}$
δ	Nernst-layer thickness	$25 \times 10^{-6} \text{ m}$
$\varepsilon, \varepsilon_s$	pore-liquid / solid- I_2 fraction	$\varepsilon = \varepsilon_0 - \varepsilon_s$
V_m	molar volume of $I_2(s)$	$5.15 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$
Q	areal capacity (charge clock)	mA h cm^{-2} ($= 3.6 \times 10^4 \text{ C m}^{-2}$)
S	super-saturation stress $c_{I_2}^f / c_{\text{sat}}$	dimensionless
S_{base}	within-cycle baseline stress	$(\Pi_{\text{inv}} + \Pi_{\text{gen}}) / \beta_{\text{intra}}$
S_{eff}	composition-resolved risk coordinate	$S_{\text{base}} \lambda_{\text{ZnI}_2}^{\text{rel}} / \lambda_{Br}^{\text{rel}}$
$\Lambda_{Br}^{\text{rel}}$	relative iodine-accommodation factor	baseline-normalized ($= 1$ at 4 M NH_4Br)
$K_{\text{sol,Br}}^{\text{app}}$	apparent capacity-level Br accommodation coeff.	$\approx 0.5 \text{ M}^{-1}$; not micro. $I_2\text{Br}^-$ const.
$\lambda_{\text{ZnI}_2}^{\text{rel}}$	relative excess-iodide stress index	baseline-normalized; weakly identified
$\theta_{\text{eff}}, A_{\text{base}}$	blocked / accessible carbon fraction	$A_{\text{base}} = 1 - \theta_{\text{eff}}$
a_s	felt specific surface area	$4 \times 10^4 \text{ m}^{-1}$
N, g_v	island areal density; deposit shape factor	m^{-2} ; $O(1)$
k_{geo}	accessibility-closure prefactor (Eq. 17)	$\pi / (a_s g_v)^{2/3}$
ε_s^*	functional-blocking solid fraction ($\theta_{\text{eff}} = \frac{1}{2}$)	closure-cond.; $\sim 5 \times 10^{-3}$
ϕ_{ppt}	fraction of generated iodine retained as solid	dimensionless; closure-cond.
$i^* = 2Fk$	dissolution-limited current	A m^{-2}

2.1 Physical picture and assumptions

We treat the positive felt as a volume-averaged porous slab ($x \in [0, L]$, collector at $x = 0$, separator at $x = L$) carrying the single faradaic couple $2I^- \rightleftharpoons I_2 + 2e^-$. The continuum scaffold (§3; SI) transports just *two* pore-phase species by reaction–diffusion: the reduced iodide $R \equiv I^-$ and a *lumped* oxidized-iodine pool O (I_3^- , $I_2\text{Br}^-$ and aqueous I_2 in fast complexation equilibrium). Because bromide is in large excess (4 M NH_4Br), ionic migration is not resolved explicitly but is folded into an effective formal potential E^{0r} in the Butler–Volmer law; iodide is a genuinely transported, depleting variable (it sinks the faradaic source), not a fixed parameter. This is a reduced two-species scaffold, not a multi-ion Nernst–Planck solve. Two ingredients carry the mechanism.

First, the neutral intermediate is *buffered*: fast complexation partitions the total dissolved iodine $c_{I_2,\text{tot}} = c_{I_2}^f + c_{I_3^-} + c_{I_2\text{Br}^-}$ instantaneously into a precipitable free monomer,

$$c_{I_2}^f = \frac{c_{I_2,\text{tot}}}{\beta}, \quad \beta = 1 + K_I c_{I^-} + K_{Br} c_{Br^-}, \quad (1)$$

so β is the factor by which complexation inflates the soluble inventory above the precipitable monomer; it falls from $\beta \approx 1263$ at full iodide toward $\beta \approx 554$ as iodide is consumed on charge (the reduction reads β at the running iodide; freezing it at the initial value is a stated approximation that shifts S but not its ordering). Second, solid blocking enters the two-electron Butler–Volmer kinetics *only* through the accessible-area prefactor $(1 - \theta_{\text{eff}})$ — the coupling that later produces the coverage–Tafel degeneracy (§4) — while $I_2(s)$ grows from supersaturated free iodine and redissolves at the Zhao rate. The pore-phase species balances, two-phase charge conservation, the Bruggeman corrections $D_{\text{eff}} = D_0 \varepsilon^{1.5}$, and the precipitation/dissolution ODE are stated explicitly as the governing balances of §2.2 (full implementation in the SI); together with the closures of §3 they form the continuum iodine-state scaffold, which we now reduce to a single surface stress that the scaffold’s averaged super-saturation reproduces up to a linear normalization. The scaffold deliberately does *not* resolve discrete pore-throat clogging, explicit multi-ion migration (a full-ion reconstruction was not numerically tractable and is unnecessary under excess bromide), the pump/pressure hydrodynamics, the collector-side shell, or the noise-level failure-state impedance; these lie outside the present reduction and enter only as bounds or external checks.

Table 2 Assumptions of the positive-electrode scaffold, the reduced stress S , and the morphology closure: what each assumes, why, its justification in the actual model, and what its failure would change. The lumped pool is $O = \{I_3^-, I_2Br^-, I_2(aq)\}$; the scaffold is the continuum solve of §3.

assumption	why used	justification (actual model)	consequence if violated
A1 Positive (iodine) felt only; the Zn negative electrode enters as a boundary.	Capacity and runaway are set on the iodine side.	The couple $2I^- \rightleftharpoons I_2 + 2e^-$ and $I_2(s)$ live only in the positive felt.	Plating-side losses add a series term not in S .
A2 Volume-averaged porous slab, $x \in [0, L]$.	Reduces a 3D felt to a through-plane scaffold.	Standard porous-electrode averaging; in-plane gradients small for the tested cell.	Strong in-plane heterogeneity breaks the 1D reduction.
A3 Fast speciation $\Rightarrow$ algebraic buffer β (Eq. 1).	One factor carries complexation instead of extra ODEs.	Complexation is fast vs the charge clock; β partitions instantaneously.	A lagging speciation makes c_L^+ (hence S) history-dependent.
A4 Oxidized iodine transported as one lumped pool O with $R = I^-$; not full multi-ion Nernst-Planck.	A two-species balance is the tractable scaffold.	SI balances solve only c_R, c_O ; per-ion migration is not reconstructed.	Per-ion migration fields are missed; S would need re-derivation from resolved fluxes.
A5 Excess Br^- ; its migration folded into a formal potential E^0 .	Removes a stiff charge-transport coupling the lumped pool cannot resolve.	Bromide is the majority anion, so its potential drop dominates and is near-constant on the iodine scale.	At low Br^- both E^0 and β drift, biasing S .
A6 Quasi-steady Nernst layer δ : $D_{\text{eff}} \Delta c_L^+ / \delta = g / \beta$.	Yields the closed surface concentration (Eq. 8) feeding Π_{gen} .	Boundary-layer relaxation is fast vs the charge clock.	A transient front makes Π_{gen} rate-history dependent.
A7 Precipitation only while super-saturated, $(S - 1)_+$; $I_2(s)$ redissolves at the Zhao rate.	Ties solid growth to the same S that sets onset.	SI solid ODE with $R_{\text{prec}} \propto (S - 1)_+$.	A nucleation barrier or non- S growth law shifts Q_s and ΔQ .
A8 ε_s is the native solid I_2 inventory (solid volume fraction), distinct from any accessibility measure.	Keeps the solved state variable separate from its post-processed consequence.	The scaffold's ε_s is a raw inventory.	Conflating inventory with blocking double-counts the closure.
A9 $\theta_{\text{eff}}(\varepsilon_s)$ (Eq. 17) is a calibrated, closure-conditioned map, not a free parameter or material constant.	Converts inventory into the area penalty $(1 - \theta_{\text{eff}})$.	3D single-fibre solve; θ_{eff} is morphology- (N -) dependent.	A different N or growth habit rescales ε_s^* and Q_s .
A10 Functional blocking defined at $\theta_{\text{eff}} = \frac{1}{2}$; rate ceiling $i^* = 2Fk$ taken as an external scale.	Fixes the two thresholds without tuning to capacity.	ε_s^* is the $\theta_{\text{eff}} = 0.5$ point; $i^* = 2Fk$ is an independent anchor.	A stricter onset shrinks ΔQ ; a wrong k rescales the rate threshold.

2.2 Governing balances of the positive-electrode scaffold

The reduction that follows distils a continuum iodine-state scaffold whose minimal closed set we state here, so that the master stress reads as a consequence of an explicit porous-electrode balance rather than an assertion; the full implementation (advective transport, reservoir and boundary conditions, parameter provenance) is given in the SI. Along the through-plane coordinate the scaffold advances the iodide c_R and the lumped oxidized pool c_O by reaction–diffusion,

$$\varepsilon_0 \partial_t c_R = \nabla \cdot (D_{R,\text{eff}} \nabla c_R) - 2R_F, \quad (2)$$

$$\varepsilon_0 \partial_t c_O = \nabla \cdot (D_{O,\text{eff}} \nabla c_O) + R_F - \frac{1}{V_m} \partial_t \varepsilon_s, \quad (3)$$

with $R_F = a_{\text{eff}} j_F / (2F)$ the volumetric faradaic rate (defined below), two iodide consumed per oxidized unit ($2I^- \rightarrow I_2 + 2e^-$) and the dissolved oxidized pool removed by net precipitation to $I_2(s)$: the net solid source $\frac{1}{V_m} \partial_t \varepsilon_s = R_{\text{prec}} - R_{\text{diss}}$ enters Eq. (3) with the opposite sign. Species storage uses the felt porosity ε_0 in the dilute-solid approximation ($\varepsilon_s \ll \varepsilon_0$), the running pore-liquid fraction $\varepsilon = \varepsilon_0 - \varepsilon_s$ entering only the transport closures (Eq. (7)). Charge is conserved in the solid and electrolyte phases,

$$\begin{aligned} \nabla \cdot \mathbf{i}_s &= -a_{\text{eff}} j_F, & \nabla \cdot \mathbf{i}_l &= +a_{\text{eff}} j_F, \\ \mathbf{i}_s &= -\sigma_{s,\text{eff}} \nabla \phi_s, & \mathbf{i}_l &= -\kappa_{l,\text{eff}} \nabla \phi_l, \end{aligned} \quad (4)$$

the interfacial rate following concentration-dependent Butler–Volmer kinetics gated by the accessible-area fraction,

$$\begin{aligned} j_F &= k_{\text{BV}} \left[c_R e^{(1-\alpha)nF\eta/RT} - c_O e^{-\alpha nF\eta/RT} \right], \\ \eta &= \phi_s - \phi_l - E^0, \quad a_{\text{eff}} = a_s (1 - \theta_{\text{eff}}), \end{aligned} \quad (5)$$

with j_F the local interfacial faradaic current density per unit accessible carbon area, k_{BV} the heterogeneous rate constant (distinct from the dissolution constant k_0 of §2.6), α the transfer coefficient, η the local overpotential, E^0 the bromide-shifted formal potential (so the volumetric faradaic current is $a_{\text{eff}} j_F = 2FR_F$ with R_F the rate of Eqs. (2)–(3)), and $\sigma_{s,\text{eff}}, \kappa_{l,\text{eff}}$ the effective solid- and electrolyte-phase conductivities of Eq. (4). The native solid $I_2(s)$ inventory grows only while super-saturated,

$$\begin{aligned} \partial_t \varepsilon_s &= V_m (R_{\text{prec}} - R_{\text{diss}}), \\ R_{\text{prec}} &\propto (S - 1)_+, \quad R_{\text{diss}} \propto (1 - S)_+ \varepsilon_s, \end{aligned} \quad (6)$$

where $S = c_{I_2}^f / c_{\text{sat}}$ is the surface super-saturation reduced to closed form in §2.3. The solid feeds back on transport only through algebraic closures,

$$\begin{aligned} \varepsilon &= \varepsilon_0 - \varepsilon_s, \quad \{D, \kappa_l\} \propto \left(\frac{\varepsilon}{\varepsilon_0} \right)^{3/2}, \\ K &= \left(\frac{\varepsilon}{\varepsilon_0} \right)^3 \left(\frac{1 - \varepsilon_0}{1 - \varepsilon} \right)^2, \quad \theta_{\text{eff}} = \theta_{\text{eff}}(\varepsilon_s), \end{aligned} \quad (7)$$

with ε_0 the felt porosity, ε the running pore-liquid fraction (Table 1), K the relative permeability, and $\theta_{\text{eff}}(\varepsilon_s)$ the accessibility closure derived from the fibre geometry (Eq. 17). The headline relation $D_{\text{eff}} = D_0 \varepsilon^{1.5}$ thus carries the full Bruggeman modifier $\varepsilon^{1.5}$ on the free diffusivity (the baseline Π_{gen} below is quoted at $D_{\text{eff}} \rightarrow D_0$), and that modifier factors as $\varepsilon^{1.5} = \varepsilon_0^{1.5} (\varepsilon / \varepsilon_0)^{3/2}$ into a felt-porosity part $\varepsilon_0^{1.5}$ that responds to *compression* and a residual $(\varepsilon / \varepsilon_0)^{3/2}$ that responds to *precipitation blockage*.

Of these, c_R, c_O, ϕ_s, ϕ_l and ε_s are solved primary fields; $\varepsilon, D, \kappa_l, K$ and θ_{eff} are algebraic closures evaluated on them; and the super-saturation S with the dimensionless groups of §2.3 is the reduced coordinate read off the solved state — the separation made explicit in the model stack (Table 3) and the implementation-to-theory map of the SI. Convection enters as a native Darcy/porous-media field whose velocity is solved and fed back into transport (full two-way coupling); in practice this feedback is negligible—the solved velocity varies by $< 0.1\%$ across a charge and a frozen-velocity control gives the identical failure onset—because precipitation barely lowers the permeability before failure (§4.7). Finally, integrating the faradaic source across the thickness under galvanostatic control gives $\frac{1}{2F} \int_0^L a_{\text{eff}} j_F dx = i_{\text{app}} / 2F$ independent of the accessibility profile, which is why the generation group Π_{gen} below is built from $i_{\text{app}} / 2F$ and carries no explicit $a_s L$ factor.

2.3 Free-I₂ balance and the super-saturation stress

At the carbon reaction front, the quasi-steady limit of the species balance (SI) across a Nernst layer of thickness δ balances the generation flux $g = i_{\text{app}}/(2F)$ against diffusive removal, $D_{\text{eff}} \Delta c_{\text{I}_2}^f / \delta = g/\beta$ across the layer; combined with Eq. (1) the surface free-iodine concentration is

$$c_{\text{I}_2}^f = \frac{1}{\beta} \left[c_{\text{I}_2, \text{tot}} + \frac{i_{\text{app}} \delta}{2F D_{\text{eff}}} \right], \quad (8)$$

and the local super-saturation is $S = c_{\text{I}_2}^f / c_{\text{sat}}$. The bracket in Eq. (8) is exactly the surface variable carried by the continuum model (Sec. 3): the first term is the accumulated oxidized-iodine inventory, the second is the generation-driven boundary-layer build-up.

2.4 Dimensionless groups and the master criterion

Dividing Eq. (8) by c_{sat} defines three dimensionless groups,

$$\Pi_{\text{gen}} = \frac{i_{\text{app}} \delta}{2F D_{\text{eff}} c_{\text{sat}}}, \quad \Pi_{\text{inv}} = \frac{c_{\text{I}_2, \text{tot}}}{c_{\text{sat}}}, \quad \beta \text{ (Eq. 1),} \quad (9)$$

so that the *within-cycle* baseline super-saturation stress is

$$S_{\text{base}} = \frac{\Pi_{\text{inv}} + \Pi_{\text{gen}}}{\beta_{\text{intra}}(Q)}, \quad \beta_{\text{intra}} \equiv \beta \text{ (Eq. 1),} \quad (10)$$

read at the running iodide; this is the coordinate the continuum model solves and that the two-threshold, compression and rate analyses below all use (throughout those analyses “ S ” denotes S_{base} at the modeled baseline electrolyte). The single buffer β_{intra} captures the within-cycle chemistry but, as the electrolyte sweeps show, cannot carry composition *across* cells (§5); the organizing coordinate is therefore the composition-resolved stress

$$S_{\text{eff}} = S_{\text{base}} \frac{\chi_{\text{ZnI}_2}^{\text{rel}}(c_{\text{ZnI}_2})}{\Lambda_{\text{Br}}^{\text{rel}}(c_{\text{Br}})}, \quad (11)$$

where the two composition factors, referred to the modeled baseline electrolyte, are

$$\Lambda_{\text{Br}}^{\text{rel}}(c_{\text{Br}}) = \frac{1 + K_{\text{sol,Br}}^{\text{app}} c_{\text{Br}}}{1 + K_{\text{sol,Br}}^{\text{app}} c_{\text{Br},0}}, \quad \chi_{\text{ZnI}_2}^{\text{rel}}(c_{\text{ZnI}_2}) = \frac{1 + \chi_I^{\text{app}} c_{\text{I}^-}}{1 + \chi_I^{\text{app}} c_{\text{I},0}}, \quad (12)$$

with baseline values $c_{\text{Br},0} = 4 \text{ M}$ and $c_{\text{I},0} = 2 \text{ M}$ iodide (1 M ZnI₂), so that $\Lambda_{\text{Br}}^{\text{rel}} = \chi_{\text{ZnI}_2}^{\text{rel}} = 1$ and $S_{\text{eff}} = S_{\text{base}}$ at baseline. Physically, $\Lambda_{\text{Br}}^{\text{rel}}$ is an apparent iodine-accommodation factor (supporting bromide raising the effective I₂ solubility) and $\chi_{\text{ZnI}_2}^{\text{rel}}$ an apparent excess-iodide stress index. Their calibration — and the sense in which $K_{\text{sol,Br}}^{\text{app}}$ is an apparent capacity-level coefficient, not the microscopic $\text{I}_2 + \text{Br}^- \rightleftharpoons$

I_2Br^- association constant — is in §5. Π_{gen} is a generation/transport Damköhler number: the current density (rate) and the effective diffusivity (and hence *compression*, through the Bruggeman relation $D_{\text{eff}} = D_0 \varepsilon^{1.5}$ [22]) enter only here. We quote the baseline reference value with the free diffusivity ($D_{\text{eff}} \rightarrow D_0$, the Nernst layer sitting in the free electrolyte), $\Pi_{\text{gen}} \approx 78$, so that compression enters as the $\varepsilon^{1.5}$ modifier rather than being folded into the baseline number. Π_{inv} carries the state of charge and the iodide inventory (ZnI₂ concentration). The buffer β falls as iodide is consumed during charge — its *within-cycle* role, which the continuum solve carries faithfully; it also rises with supporting bromide, but only weakly ($K_{\text{Br}c_{\text{Br}}} \ll K_{\text{I}c_{\text{I}^-}}$), so the large *cross-composition* bromide effect measured in full cells is not carried by β but by a separate apparent iodine-accommodation term (§5). Equation (11) is the composition-extended organizing coordinate: it separates the within-cycle stress S_{base} from cross-composition iodine accommodation and excess-iodide stress, in the spirit of Damköhler-type dimensionless reductions of flow-cell behaviour [23]. At the modeled baseline, precipitation onset is $S_{\text{base}} = 1$, where free I₂ first saturates. For composition sweeps, S_{eff} serves as a capacity-level risk coordinate normalized to that same threshold, the accommodation factors $\Lambda_{\text{Br}}^{\text{rel}}$ and $\chi_{\text{ZnI}_2}^{\text{rel}}$ (Eq. 11) extending the baseline reference across composition rather than asserting a measured solubility at each composition.

2.5 Two-threshold passivation: retained versus blocking

Precipitation onset does not coincide with electrode failure. We define two thresholds:

1. **Baseline precipitation onset**, $S_{\text{base}} = 1$, reached at areal capacity Q_s (for composition sweeps S_{eff} is normalized to this same risk threshold), where free I₂ first saturates (c_{sat} being the measured aqueous I₂ solubility [24]) and solid I₂(s) begins to nucleate.
2. **Functional blocking**, $\varepsilon_s = \varepsilon_s^*$, reached at Q_f , where the solid fraction crosses the accessibility/percolation threshold and the accessible reaction area collapses — a non-conducting solid deposit blocking a porous electrode by percolation has precedent in porous-electrode modeling [25] and in mineral-clogging/percolation models of porous media [26].

This functional blocking applies to *unconfined* bulk I₂(s) in macroporous flow-cell electrodes; nanoconfined I₂(s) in microporous carbons can instead remain electrochemically recoverable [13, 14], so it is the pore-scale geometry — not precipitation *per se* — that decides whether retained iodine becomes blocking iodine. Between them the solid accumulates only while super-saturated. We track ε_s as the deposited-I₂(s)

volume fraction on the *total-electrode-volume* basis throughout (consistent with the continuum closure $\varepsilon = \varepsilon_0 - \varepsilon_s$ and the SI), so against the charge passed per electrode area $Q_C = \int i_{\text{app}} dt$,

$$\frac{d\varepsilon_s}{dQ_C} = \frac{V_m}{2FL} \phi_{\text{ppt}} (S - 1)_+, \quad (13)$$

where V_m is the molar volume of $\text{I}_2(\text{s})$, L the electrode thickness, ϕ_{ppt} the precipitating fraction of the generated iodine, $(\cdot)_+$ the positive part, and $Q_C = 3.6 \times 10^4 Q$ for Q in mAh cm^{-2} . A pore-volume fraction, where the fibre closure needs one, is the rescaling $\hat{\varepsilon}_s = \varepsilon_s/\varepsilon$. Compression therefore enters not the accumulation slope but the *threshold*: functional blocking is a fixed pore occupancy at percolation $\hat{\varepsilon}_s^*$, i.e. a total-volume threshold $\varepsilon_s^* = \varepsilon \hat{\varepsilon}_s^* \propto \varepsilon$, so in the clean deep-fill limit $Q_f = \varepsilon_s^* 2FL/[V_m \phi_{\text{ppt}}(S - 1)_+] \propto \varepsilon L$, recovering the compression scaling used below. The accessible area follows the geometric closure $a_{\text{eff}} = a_s A_{\text{bare}}$ with $A_{\text{bare}} = 1 - \theta_{\text{eff}}(\varepsilon_s)$, θ_{eff} being fixed by the mesoscale model of Sec. 3. Failure occurs at $\varepsilon_s(Q_f) = \varepsilon_s^*$. The gap $\Delta Q = Q_f - Q_s > 0$ is the capacity over which iodine is retained but not yet blocking; it is the quantitative content of “retained iodine is not blocking iodine.”

2.6 Dissolution-limited current and the rate threshold

At steady state with solid present, the largest current that the dissolution can sustain without runaway follows from $g_{\text{diss}} = k$, giving the dissolution-limited current

$$i^* = nFk, \quad n = 2, \quad (14)$$

the form measured by Zhao *et al.* [16]; a transport-limited current of this $i^* = nFk$ form is the established ceiling for porous flow electrodes [27]. If the dissolution capacity is limited by a second-order iodide-dependent recovery rate, $k \simeq k_0(c_{\text{I}}/c_{\text{I}}^0)^2$, and iodide depletes linearly with charge, $c_{\text{I}}(Q) = c_{\text{I}}^0(1 - Q/Q_{\text{iod}})$, then the Damköhler number rises as $\text{Da}(Q) = i_{\text{app}}/i^*(Q) = \text{Da}_0(1 - Q/Q_{\text{iod}})^{-2}$ ($\text{Da}_0 = i_{\text{app}}/i_0^*$, $i_0^* = 2Fk_0$) and runaway ($\text{Da} = 1$) occurs at

$$\frac{Q_f}{Q_{\text{iod}}} = 1 - \sqrt{\text{Da}_0} = 1 - \sqrt{i_{\text{app}}/i_0^*}. \quad (15)$$

We use Eq. (15) as an illustrative order-of-magnitude kinetic ceiling, not as an independently identified rate law for the present electrolyte: it inherits the assumed second-order iodide dependence (the order Zhao *et al.* measure is in I_2), while the robust external input is the dissolution-limited scale $i^* = 2Fk$ itself.

2.7 Failure-onset scaling

Two regimes follow from Eqs. (13)–(15). At low rate ($\text{Da}_0 \ll 1$) failure is set by the geometric accommodation to ε_s^* , and

in the clean deep-fill limit the failure capacity scales with the *pore volume per unit area*, $Q_f \propto \varepsilon L$ (through the percolation threshold $\varepsilon_s^* \propto \varepsilon$, Eq. 13; compression; the half-blocking onset itself is reached earlier and is far less compression-sensitive, see Mechanical compression). At high rate failure is kinetic, set by the dissolution threshold $i_{\text{app}} = i_0^*$ (rate). Supporting bromide enters chiefly through an apparent iodine-accommodation term (raising the effective I_2 solubility) rather than through the complexation buffer β , which under-predicts its measured effect; the parallel I_2Br^- dissolution channel is one of several effects that term lumps (§5). The two limits are the geometric and kinetic faces of the two-stage criterion of Eq. (11) (precipitation at $S=1$, functional blocking at $\theta_{\text{eff}}=\theta_{\text{eff}}^*$).

2.8 Cross-scale priors and closures

The baseline scaffold and morphology coefficients are not free fit parameters; they are fixed by lower-scale calculations or branch-defined physical inputs (the cross-composition factors $\Lambda_{\text{Br}}^{\text{rel}}$ and $\chi_{\text{ZnI}_2}^{\text{rel}}$ are, by contrast, separately calibrated capacity-level chemistry anchors of §5, not microscopic constants), so the theory inherits these numbers from molecular and mesoscale physics rather than from fitting, in the derive-closures-then-upscale spirit of multiscale porous-electrode theory [28] (Fig. 2). The molecular layer (DFT site/coalescence priors and MD carrier diffusivities) and the mesoscale layer (a three-dimensional fibre accessibility model and a pore-throat network) are reported with their results in Sec. 4 (Figs. 3–5); here we record the closure they hand to Eq. (13). The fibre model resolves $\text{I}_2(\text{s})$ as discrete surface islands of areal number density N and radius r , whose volume per electrode volume and random-overlap projected coverage are

$$\varepsilon_s = a_s N g_v r^3, \quad \theta_{\text{geo}} = 1 - \exp(-\pi N r^2), \quad (16)$$

with $g_v = O(1)$ the deposit shape factor ($g_v = \frac{4}{3}\pi$ for a sphere). Eliminating the unobserved radius, $r = [\varepsilon_s/(a_s N g_v)]^{1/3}$, makes the accessibility loss depend on the solid fraction through a fixed 2/3 power,

$$\theta_{\text{eff}}(\varepsilon_s) = 1 - \exp(-k_{\text{geo}} N^{1/3} \varepsilon_s^{2/3}), \quad k_{\text{geo}} = \frac{\pi}{(a_s g_v)^{2/3}}, \quad (17)$$

so the exponent is geometric, not empirical; the fibre accessibility solve reproduces both powers (on N , 0.33 vs. 1/3; on ε_s , 0.63 vs. 2/3), and its calibrated prefactor at the dense-nucleation case corresponds to a physical $O(1)$ island shape factor. The functional-blocking fraction ε_s^* is its $\theta_{\text{eff}} = \frac{1}{2}$ point and the relative permeability $K(\varepsilon_s)$ is supplied by the network. These closures, not adjustable parameters, are what

make Eqs. (1)–(15) a closed-form, closure-conditioned reduction.

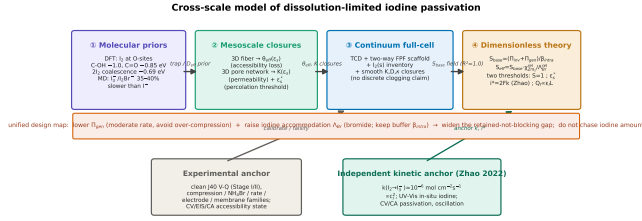


Figure 2 Cross-scale model. Molecular priors (DFT trapping, MD transport) set the closures; 3D fiber/pore-network models supply the accessibility $\theta_{\text{eff}}(\varepsilon_s)$ and permeability $K(\varepsilon_s)$ closures and accessibility/permeability cross-checks; the continuum model integrates them using a calibrated, closure-conditioned blocking threshold ε_s^* , and the reduced stress framework distils the model stack into design rules. The experimental families and the independent dissolution kinetics provide external anchors and scope checks for the chain.

3 Computational model

No single layer suffices, and conflating layers is the main hazard of a cross-scale study. The continuum solve generates the positive-electrode *state* fields, but its native solid fraction is not itself an accessibility measure and its raw galvanostatic voltage is weaker than experiment; the molecular layers *bound* parameters but supply no rates; the mesoscale models *convert* solid inventory into accessibility and permeability consequences; the reduced layer only *reads* the voltage; and the experiments are external *anchors*. Table 3 states each layer with the question it answers, what it passes on, and its claim level, so that solved fields, postprocessed closures, bounded priors and external anchors are never conflated.

Table 3 The model stack. Each layer answers a distinct question and is labelled by claim level; arrows between them carry the variables named in the third column (cf. Fig. 2).

layer	question it answers	output passed on	claim level
molecular (DFT, MD)	where/how strongly is I_2 retained; how fast do carriers move?	site/coalescence bias; bounded D_{eff}	prior (bound, not a rate)
dimensionless theory (§2)	when does free I_2 supersaturate, and when does it block?	$S, Q_s, Q_i, \Delta Q$	derived closed form
mesoscale closures (fibre, pore network)	when does retained $I_2(s)$ become inaccessible/flow-limiting?	$\theta_{\text{eff}}(\varepsilon_s), \varepsilon_s^*, K(\varepsilon_s)$	calibrated closure
continuum solve (COMSOL)	what are the 2D state fields in a real cell?	$S, \varepsilon_s, \theta_{\text{eff}}, A_{\text{bare}}, \text{potentials}$	solved state generator
reduced voltage (Eq. 18)	does the Stage I/II curve track the state clock?	Nernst/depletion/access. split	interpretive observable
experiments (full cell: CV/CA/EIS)	what must the chain stay consistent with?	operating regime, voltage scale, knob trends	external anchor

The dimensionless theory is tested against, and its closures integrated by, a native full-cell continuum solve. A tertiary-current-distribution model [25, 29–31] solves the governing balances of §2.2 (Eqs. (2)–(7)) in the real cell geometry, with native two-way Darcy/porous-media flow — the ve-

locity solved and coupled back into transport, the permeability feedback mild within the analysed window (§4.7): galvanostatic Butler–Volmer kinetics for $2I^- \rightarrow I_2 + 2e^-$ with a Nernstian equilibrium potential, the surface supersaturation S of Eq. (11), a precipitation/dissolution source $d\varepsilon_s/dt = V_m(R_{\text{prec}} - R_{\text{diss}})$ with $R_{\text{prec}} \propto (S - 1)_+$ (a chemical-potential-driven phase-transfer source of the kind standard in porous-electrode continuum theory [32]), and the accessibility, permeability and transport closures of Sec. 2.8. The reported cell voltage is the solved galvanostatic response, not an empirical fit; numerical details are in the Methods. For *interpreting* the voltage we additionally use a Butler–Volmer reduction,

$$V(Q) = V_0 + \underbrace{\frac{RT}{2F} \ln c_{I_2}^f}_{\text{Nernst}} + b_T \left[\underbrace{\ln \frac{c_{I^-}^0}{c_{I^-}(Q)}}_{\text{iodide depletion}} + \underbrace{\ln \frac{1}{A_{\text{bare}}(Q)}}_{\text{accessibility}} \right] + i_{\text{app}} R_{\Omega}, \quad (18)$$

fitted only in its physical kinetic parameters (V_0 and an apparent Tafel slope b_T).

4 Results

We present the results bottom-up because the model is built from priors and closures into a continuum state generator; the scientific question, however, is problem-driven: *when does soluble or retained iodine become functionally blocking?* We therefore first establish the molecular and transport bounds that fix where iodine is retained and how fast it clears (§4.1); then use the mesoscale closures that convert retained solid $I_2(s)$ into accessibility and permeability consequences (§4.2); and finally test whether the resulting positive-electrode state clock—through the continuum solve and the dimensionless stress that integrate and distil them—explains the observed transition from normal operation to the voltage rise, across voltage, compression and operating-knob anchors.

4.1 Molecular priors: where iodine is retained and how fast it clears

The chain begins at the molecular scale, which supplies *bounds and tendencies*—not rate constants—for two quantities the upper layers inherit. Periodic CP2K places molecular I_2 on the carbon surface with single- I_2 adsorption energies of -0.40 (basal), -0.85 (carbonyl, $C=O$) and -1.00 eV (hydroxyl, $C-OH$), and -0.32 eV at a vacancy: oxygenated defects bind I_2 roughly twice as strongly as pristine basal, and a two-molecule coalescence of -0.69 eV at $C-OH$ favours compact iodine clusters (Fig. 3a; charge-density-difference

maps and projected densities of states in the SI). These set the *tendency* to retain and locally co-associate iodine at oxygenated sites—a placement bias, not a rate—consistent with independent DFT of iodine on carbon [33, 34]. A single-point speciation proxy is direction-only: the triiodide branch is clearly downhill ($\Delta E = -0.46$ eV), justifying retention of the I_2/I_3^- constant K_I as a calibrated macroscopic parameter, whereas the bromide branch is borderline-uphill (+0.05 eV) and enters only as a weak term (Fig. 3b).

Classical molecular dynamics then bounds how quickly oxidised iodine clears. The self-diffusivities form a clear carrier ladder (Fig. 3c): H_2O and NH_4^+ fastest, reduced I^- mid-ladder (1.22 ± 0.03), and the oxidised carriers systematically slower— I_3^- 0.75 ± 0.07 and I_2Br^- 0.72 ± 0.09 , a 38–41% reduction ($\times 10^{-9} m^2 s^{-1}$). Two results pass upward: oxidised iodine is intrinsically $\sim 40\%$ less mobile than iodide—the removal lag that seeds local accumulation and bounds D_{eff} in Π_{gen} —and the low-end carrier-mix window (~ 0.4 – $0.7 \times 10^{-9} m^2 s^{-1}$) is the bounded effective diffusivity the mesoscale and continuum layers adopt for the iodine-laden film. These bounded priors do not yet block anything; converting the resulting solid inventory into a transport penalty is the task of the mesoscale closures.

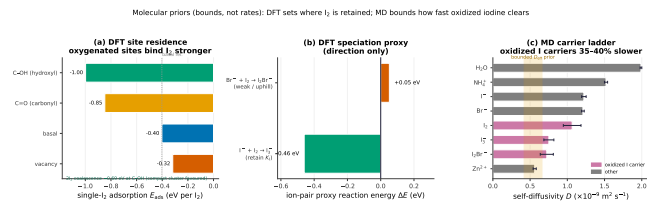


Figure 3 Molecular priors (bounds, not rates). (a) DFT single- I_2 adsorption energies (basal -0.40 , carbonyl -0.85 , hydroxyl -1.00 , vacancy -0.32 eV): oxygenated defects bind I_2 roughly twice as strongly as basal, and a $2I_2$ coalescence of -0.69 eV at C–OH favours compact clusters. (b) DFT speciation proxy (direction only): the I_3^- branch is downhill, the I_2Br^- branch borderline-uphill. (c) MD carrier diffusivities: the oxidised carriers I_3^- and I_2Br^- are 38–41% slower than I^- (neutral I_2 only $\sim 12\%$); shaded band is the low-end D_{eff} prior.

4.2 Mesoscale closures: from retained solid to functional blocking

The mesoscale layer converts the continuum solid inventory ε_s into the two transport closures the theory needs (the two structures are rendered in Fig. 4). A battery-calibrated three-dimensional single-fibre deposition model ($r_f = 7.5 \mu m$; Fig. 4a) gives the accessibility loss $\theta_{eff}(\varepsilon_s)$ of Eq. (17) [35] (Fig. 5a): it lies *below* the raw geometric coverage at every loading ($\theta_{geo} = 0.14 \rightarrow 0.52$ versus $\theta_{eff} = 0.11 \rightarrow 0.42$ over $\varepsilon_s = 3.4 \times 10^{-4}$ – 3.4×10^{-3}) because three-dimensional access partly relieves the block, and the retained accessibility declines *smoothly* from 0.89 to 0.58—about 40% of the sur-

face lost near end of charge, with no abrupt passivating film. The functional-blocking threshold ε_s^* is set by spatial percolation and depends on nucleation density ($\varepsilon_s^* = 2.2 \times 10^{-3}$ at $10^{14} m^{-2}$, 6.8×10^{-3} at 10^{13} , never reached at 10^{11}); the operating loading $\varepsilon_s \sim 2$ – 3×10^{-3} sits right at this onset edge.

A pore-throat network (Fig. 4b) then tests whether precipitated I_2 throttles transport by clogging. Sweeping six deposit-placement laws, at the operating load ($\varepsilon_s \approx 3.8 \times 10^{-3}$) the network stays 98–99% permeable for *every* law and the active backbone fully percolates (Fig. 5b); halving the permeability requires 30–100 $\times$ the operating load, so bulk clogging cannot explain the accessibility loss. The network does supply one non-continuum constraint a Kozeny–Carman law misses: blocking efficiency is set by the spatial *overlap* of deposit, not its amount (path-correlated deposition blocks $\sim 1.8\times$ more efficiently than surface-spread at the same loading; Fig. 5c). With $\theta_{eff}(\varepsilon_s)$, its threshold ε_s^* , and the near-unity $K(\varepsilon_s)$ now fixed by lower-scale physics rather than fitting, the continuum model integrates them in a real cell, and the dimensionless theory below distills the solved super-saturation field.

Mesoscale models behind the closures: the calibrated structures whose $\theta_{eff}(\varepsilon_s)$ and $K(\varepsilon_s)$ enter the continuum solve

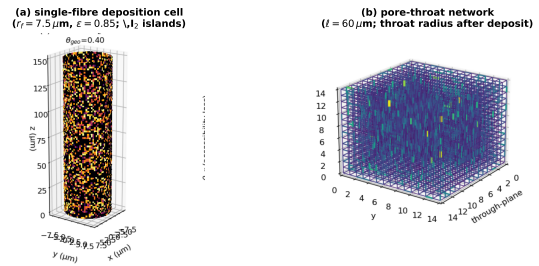


Figure 4 The mesoscale models. (a) The battery-calibrated single-fibre deposition cell ($r_f = 7.5 \mu m$, $\varepsilon = 0.85$) with stochastically nucleated I_2 islands, and (b) the pore-throat network after deposit, whose accessibility $\theta_{eff}(\varepsilon_s)$ and permeability $K(\varepsilon_s)$ closures (Fig. 5) are handed to the continuum solve. At the operating solid load the deposit remains sub-monolayer islands rather than a continuous film, consistent with the percolation-limited accessibility loss.

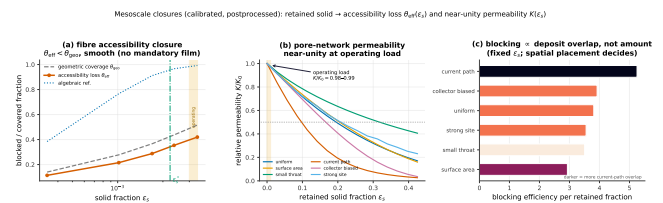


Figure 5 Mesoscale closures (calibrated, postprocessed). (a) Fibre accessibility closure $\theta_{eff}(\varepsilon_s)$ lies below the geometric coverage and declines smoothly (no mandatory film); ε_s^* is the percolation threshold. (b) Pore-network relative permeability $K(\varepsilon_s)$ for six deposit-placement laws stays near unity at the operating load; halving it needs 30–100 $\times$ more solid. (c) Blocking efficiency is set by the spatial overlap of deposit, not its amount.

4.3 The theory reproduces the model's super-saturation field

The closed-form stress of Eq. (11), evaluated with the continuum inventory, iodide and buffer fields, reproduces—as an internal-consistency check rather than a validation of the solver—the model super-saturation to $R^2 = 1.000$ up to a single constant of proportionality (slope 3.00, the solver referencing the free-monomer saturation $c_{\text{sat}}/3$ rather than the total-iodine reference; Fig. 8b): the dimensionless theory is the solver's exported super-saturation channel. The baseline generation group, evaluated with the free diffusivity D_0 as the reference state (so that compression enters as the $D_{\text{eff}} = D_0 e^{1.5}$ modifier of Eq. (11)), is $\Pi_{\text{gen}} = 77.9$, comparable to the initial surface inventory $\Pi_{\text{inv}}(0) = 78.0$. Over the charge the total oxidized-iodine inventory rises $4.16\times$ while the *free*, precipitable fraction rises $9.48\times$, because iodide is consumed 58% ($c_{\text{I}^-} : 1688 \rightarrow 704 \text{ mol m}^{-3}$) and the buffer collapses 56% ($\beta : 1263 \rightarrow 554$); the speciation identity $c_{\text{I}_2}^{\text{f}} = c_{\text{I}_2, \text{tot}}/\beta$ holds to $R^2 = 0.9999998$, so the reduction is a faithful compression of the exported super-saturation channel, not a fit. The two thresholds are clearly separated (Fig. 8a): precipitation onset $S = 1$ at $Q_s = 83 \text{ mAh cm}^{-2}$ and functional blocking at $Q_f = 100 \text{ mAh cm}^{-2}$ (defined at half-blocking $\theta_{\text{eff}} = 0.5$; an onset criterion $\theta_{\text{eff}} \geq 0.01$ gives $Q_f = 85$, so the reservoir is threshold-dependent and is computed under the accessibility closure), a retained-but-not-blocking gap of $\Delta Q \approx 17 \text{ mAh cm}^{-2}$. The full event sequence—free- I_2 super-saturation of the averaged field ($S=1$ at $Q=83$), the solver's first finitely-resolved $\text{I}_2(\text{s})$ ($Q \approx 98$, a local detection threshold above the averaged onset), accessibility loss $\theta_{\text{eff}} = 0.5/0.9$ ($99.6/111.2$) and the dV/dQ peak (116.7)—is ordered as the theory requires, with oxidized iodine appearing well before the electrode passivates. The native two-dimensional field maps (Fig. 7) and the through-plane space-time evolution (Fig. 6) make this concrete: solid I_2 and the total current localize in a thin band at one electrode face while the free- I_2 stress remains a smooth through-plane gradient, and the passivation front builds at that face with capacity.

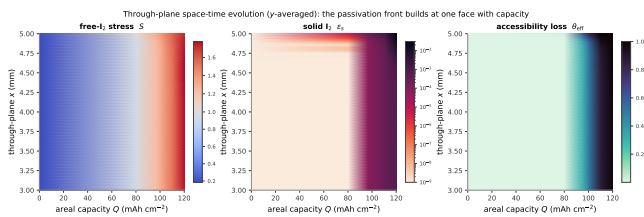


Figure 6 Through-plane space-time evolution (y-averaged): the passivation front (free- I_2 stress, solid I_2 , accessibility loss) builds at one face as capacity increases.

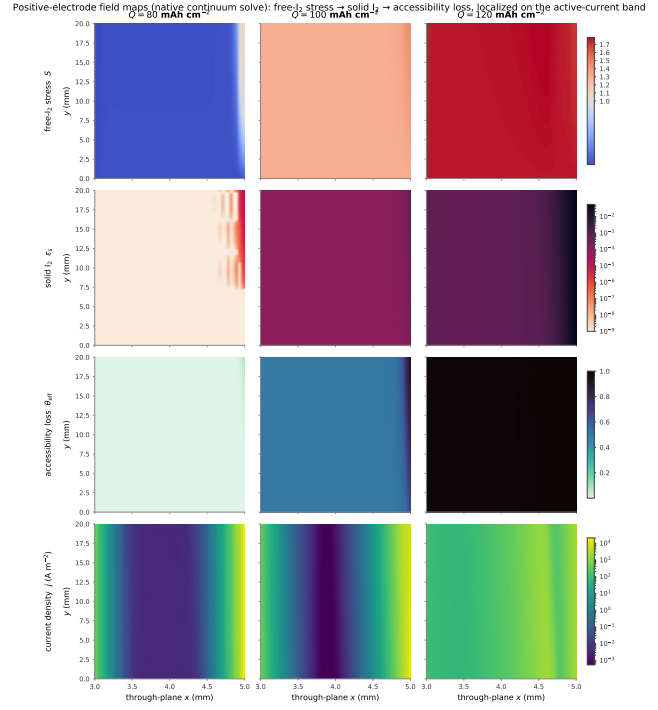


Figure 7 Native two-dimensional field maps of the positive electrode (free- I_2 stress S , solid I_2 ϵ_s , accessibility loss θ_{eff} , total current density j) at $Q = 80/100/120 \text{ mAh cm}^{-2}$. Solid iodine and current localize in a thin band at one face while S stays a smooth through-plane gradient.

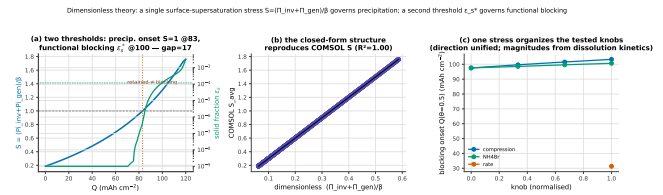


Figure 8 Dimensionless theory of dissolution-limited passivation. (a) The super-saturation stress $S_{\text{base}} = (\Pi_{\text{inv}} + \Pi_{\text{gen}})/\beta_{\text{intra}}$ crosses unity (precipitation onset, Q_s) well before the solid fraction reaches the functional-blocking threshold ϵ_s^* (Q_f); the gap is the retained-but-not-blocking capacity. (b) The closed form reproduces the continuum-model super-saturation field ($R^2 = 1.000$, up to a constant normalization). (c) The stress coordinate organizes the tested operating/design knobs by the mechanism node each perturbs.

4.4 Retained iodine co-locates with current until accessibility collapses

Post-processing the solved baseline for the partial currents (productive bare-carbon $j_{\text{bare}} = j_{\text{F}} A_{\text{bare}}$ versus precipitate-covered $j_{\text{I}_2(\text{s})}$, computed from the solved reaction terms; Fig. 7) gives the cleanest node-level test of “retained $\neq$ blocking.” The field maps are visual guides only; the claim rests on the partial-current correlation statistics and current-fraction metrics below, not on visual inspection of colour bands. Three timelines decouple within one fill: retained

iodine rises monotonically and precipitation becomes ubiquitous early (non-zero covered current at every node by $Q = 96 \text{ mAh cm}^{-2}$, so $Q_s < 96$), yet the spatial correlation between retained iodine and productive current stays strongly *positive* up to $Q = 108 \text{ mAh cm}^{-2}$ ($r(\varepsilon_s, j_{\text{bare}}) = +0.65, +0.87, +0.90, +0.90$ at $Q = 80, 96, 100, 108$), with 75–97% of the productive current carried in the iodine-rich half of the electrode (rising to 87–97% over $Q = 96$ –108) — iodine marks the active, fast-kinetics regions rather than blocking them. Blocking is a late spatial bifurcation: as the accessible area collapses $\sim 40\times$ (a percolation-type collapse analogous to mineral pore-clogging [36]) and coverage saturates ($\theta_{\text{eff}} > 0.95$ over 94% of nodes at $Q = 120$), the correlation crosses zero (+0.03 at $Q = 115$) and inverts (-0.07 at $Q = 120$), where 99% of the remaining productive current is funnelled onto the rare iodine-poor surviving islands and the early near-perfect $r(\varepsilon_s, A_{\text{bare}}) = -0.989$ degrades to -0.13 . The voltage symptom is correspondingly downstream: 84% of the total charge-voltage rise occurs only *after* half the surface is already blocked.

4.5 Dissolution-limited magnitude and independent kinetic anchor

The theory carries two quantitative magnitudes. The *geometric* face predicts that the low-rate failure capacity scales with the pore volume per area, $Q_f \propto \varepsilon L$ (Eq. 13, via the percolation threshold $\varepsilon_s^* \propto \varepsilon$); this is consistent with our full-cell compression series, reported separately, used here only as an external check on the scaling (Fig. 9a). The *kinetic* face predicts a rate ceiling at $i_{\text{app}} = i_0^* = 2Fk_0$: with the independently measured $k_0 \sim (1.2\text{--}2.2) \times 10^{-6} \text{ mol cm}^{-2} \text{ s}^{-1}$ from the in-situ iodine kinetics of Zhao *et al.* [16], this gives $i^* = 232\text{--}425 \text{ mA cm}^{-2}$, the natural dissolution-limited rate ceiling (Fig. 9b). The kinetic anchor is from an independent laboratory and electrolyte, so the magnitude scale is not fitted; it is consistent with the dissolution-limited super-saturation, not the retained amount, setting failure. Strikingly, two unrelated external anchors converge on the same ceiling: our separately-measured full-cell rate threshold (280–320 mA cm^{-2}) falls squarely inside the independent Zhao kinetic ceiling (232–425 mA cm^{-2}) — a coincidence the dissolution-limited picture predicts, but to which neither anchor was tuned.

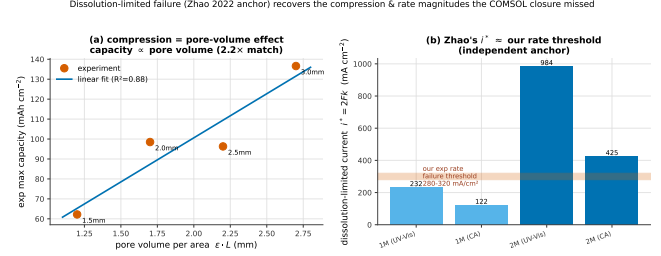


Figure 9 Dissolution-limited magnitude, externally anchored. (a) The geometric limit $Q_f \propto \varepsilon L$; full-cell capacities (reported separately) are shown only as an external consistency check. (b) The dissolution-limited current $i^* = 2Fk$ for the rate constants of Zhao *et al.* [16]; the bars span the reported k at the two iodide concentrations and the two estimation methods, and the body uses the $i^* \approx 232\text{--}425 \text{ mA cm}^{-2}$ range from $k = (1.2\text{--}2.2) \times 10^{-6} \text{ mol cm}^{-2} \text{ s}^{-1}$ (we are at 2 M I⁻).

4.6 Stage-I/II voltage: a reduced observable layer, and why no coalesced film is required

The reduced model of Eq. (18) fits the real pre-failure charge curve (plateau plus the onset of the rise) to a root-mean-square error of 6.5 mV (Fig. 10a). Decomposing the fitted response, the rise is dominated by the iodide-depletion concentration term $b_T \ln(c_T^0/c_T^-)$, with a smaller Nernstian free-I₂ contribution ($\approx 23 \text{ mV}$) and an accessibility term $b_T \ln(1/A_{\text{bare}})$ that engages only as the upturn begins (the latter two are not simply additive into the plateau-to-onset rise because each is referenced to a different point of the charge); an ohmic offset $i_{\text{app}}R_\Omega$ completes Eq. (18). The rise is therefore dominated by the cell running short of accessible iodide, with accessibility loss engaging only at the onset of the upturn. Crucially, the absolute charge voltage cannot uniquely determine the coverage morphology: the accessibility contribution to the rise is reproduced by an end-of-charge coverage anywhere from $\theta_{\text{eff}} \approx 0.6$ to 0.9 by trading off the (apparent) Tafel slope, and a near-complete coalesced film ($\theta_{\text{eff}} \rightarrow 0.99$) would require an unphysically small slope. The conventional picture of a continuous passivating iodine film [37] is thus not demanded by the data: such a “film required” conclusion is an artifact of an inflated voltage target; against the real charge curve, single-fiber-scale accessibility loss suffices, and morphology must be inferred from geometry and percolation, not from voltage. Replacing the empirical accessibility closure by the physical mesoscale closure removes the late over-blocking and the spurious voltage overshoot (Fig. 10b).

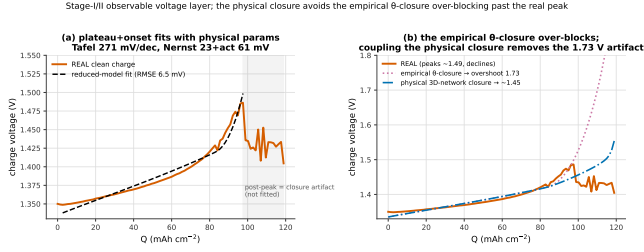


Figure 10 Stage-I/II voltage observable layer. A reduced observable layer (not the native COMSOL terminal voltage) aligns the normal and rising portions of the clean charge curve with the state clock. (a) The reduced Butler–Volmer model (Eq. 18) fits the real pre-failure charge curve to 6.5 mV; the rise is dominated by iodide depletion. (b) The empirical accessibility closure over-blocks past the real peak (voltage overshoot); coupling the physical mesoscale closure removes the artifact—no coalesced film is required.

4.7 Bulk permeability loss is small within the smooth network closure

Injecting the physically derived pore-network permeability $K(\varepsilon_s)$ in place of the mean-field Kozeny–Carman closure and re-solving the continuum model shifts the cell voltage by only 1.27 mV (Supplementary Fig. S6); the relative permeability stays near unity throughout (Kozeny–Carman endpoint 0.96, pore-network endpoint 0.98). Likewise the internal overpotential decomposition shows that the late voltage response is carried by the electronic/contact and accessibility terms, not by bulk hydraulic clogging. Failure is loss of *accessible reaction sites*, consistent with a slow pore/interface impedance state in the *pre-failure* regime (not solution resistance; the noisy failure-regime spectra are not interpreted, SI) and with the dissolution-limited mechanism. This bounds only the homogeneous, smoothly-distributed permeability penalty; it does not exclude discrete or local pore-throat blockage, collector-side shell formation, or post-onset hydraulic/morphological failure in real felt under heterogeneous compression.

This near-unity permeability is also what makes the flow coupling immaterial to failure. The continuum solve carries *full two-way* Darcy/porous-media flow (the velocity is solved and fed back into the species transport); solving it this way leaves the failure onset unchanged and the velocity field essentially frozen ($< 0.1\%$ over a charge), because K/K_0 stays at unity until the accessibility threshold is already crossed ($K/K_0 = 0.998$ at $\theta_{\text{eff}} = \frac{1}{2}$), and the pressure rises by only a few percent through the analysed window. Far past failure the feedback is no longer benign: as the pore network fully clogs ($K/K_0 \rightarrow 0$, reaching ≈ 0.25 in the most diffusion-limited case at $\theta_{\text{eff}} \rightarrow 1$) the Darcy pressure diverges super-linearly, rising $\sim 10^3\times$ toward ~ 8 bar in the frozen-throughput limit — the regime in which constant-flow pumping mechanically

ruptures tubing and seals, as observed experimentally. That fully-destabilised hydraulic runaway is a distinct, *deep-post-failure* mode downstream of the accessibility-collapse failure analysed here, and lies outside the present continuum scope.

5 Mechanistic analysis of operating and design knobs

The operating and design variables examined here act on positive-electrode iodine passivation through the dimensionless stress of Eqs. (10)–(11), baseline $S_{\text{base}} = (\Pi_{\text{inv}} + \Pi_{\text{gen}})/\beta_{\text{intra}}$ (Sec. 2), with $\Pi_{\text{gen}} = i_{\text{app}}\delta/(2FD_{\text{eff}}c_{\text{sat}})$, $\Pi_{\text{inv}} = c_{\text{I}_2,\text{tot}}/c_{\text{sat}}$, $\beta = 1 + K_{\text{I}}c_{\text{I}^-} + K_{\text{Br}}c_{\text{Br}^-}$ and the Bruggeman diffusivity $D_{\text{eff}} = D_0\varepsilon^{1.5}$. This projects the tested fixed- c_{sat} scan families onto a single *phase plane* (Fig. 11): each knob is a distinct *move* in the $(\Pi_{\text{gen}}, \Pi_{\text{inv}})$ plane relative to one precipitation-onset boundary. Because the buffer at onset is nearly constant across cells ($\beta \approx 760$), that boundary is an almost fixed budget line, $\Pi_{\text{gen}} + \Pi_{\text{inv}} \approx 253$, and the fixed- c_{sat} scan families fall on this budget line within the solved branch: rate, diffusivity and compression slide the operating point along Π_{gen} , while state of charge climbs it vertically in Π_{inv} until it crosses — the buffer falling from ≈ 1263 to 554 *along* this charge trajectory is what drives S upward. Two knobs instead move the boundary: salting-out, which rescales c_{sat} and shifts the whole plane, and — weakly within a fixed electrolyte — the within-cycle buffer β_{intra} , whose dominant cross-composition counterpart (supporting bromide) instead enters through $\Lambda_{\text{Br}}^{\text{rel}}$ (§5). Conductivity and flow move the point not at all: they are the negative control made geometric. Precipitation begins when the point crosses the onset boundary ($S = 1$, onset capacity $Q_{S=1}$); functional blocking occurs a finite distance further out, at $\theta_{\text{eff}} = 0.5$ (blocking capacity $Q_{\theta=0.5}$), and the strip between the two boundaries is the quantitative “retained $\neq$ blocking” reservoir. The accessible fraction is $A_{\text{bare}} = 1 - \theta_{\text{eff}}$, the dissolution-limited current ceiling is $i^* = 2Fk$ [16], and the failure capacity scales with the precipitate reservoir εL . We take the knobs in the order they enter the plane: generation-side transport and rate (Π_{gen}), then compression (which couples Π_{gen} to the pore-volume capacity), inventory (Π_{inv}), bromide (the buffer β), specific area, and finally the orthogonal conductivity/flow control. All values quoted are model outputs from the native COMSOL parameter scan and the associated dimensionless-theory tables.

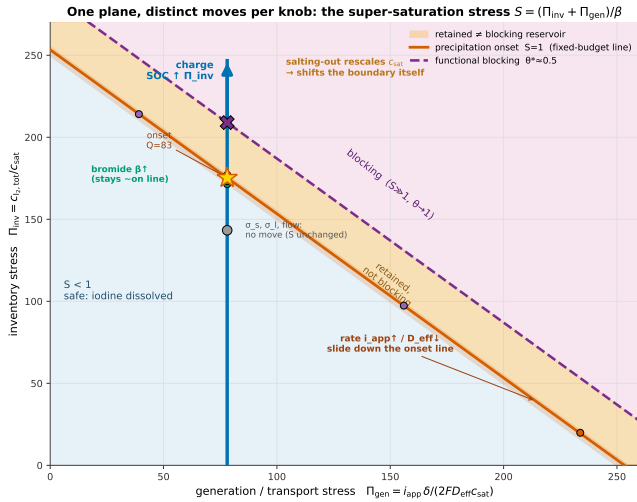


Figure 11 One plane, distinct moves per knob. The dimensionless knob phase plane. Axes are the generation/transport stress $\Pi_{\text{gen}} = i_{\text{app}}\delta/(2FD_{\text{eff}}c_{\text{sat}})$ and the inventory stress $\Pi_{\text{inv}} = c_{\text{L,tot}}/c_{\text{sat}}$, read directly from the solved surface field (at $Q = 0$ the oxidized inventory vanishes, so the surface total fixes Π_{gen}). The orange line is the precipitation onset where the solved free monomer saturates ($S = 1$); it is a nearly fixed budget line, $\Pi_{\text{gen}} + \Pi_{\text{inv}} \approx 253$, because the buffer at onset is almost constant across cells ($\beta \approx 760$), shown as a thin band for that small spread. (The larger β collapse $1263 \rightarrow 554$ occurs along a charge trajectory and is what lifts S to onset; it is not the width of the boundary.) The dashed line is functional blocking ($\theta_{\text{eff}} = 0.5$) and the amber strip between is the retained-not-blocking reservoir ($\Delta Q \approx 17 \text{ mA h cm}^{-2}$). The blue arrow is the baseline charge trajectory, climbing in Π_{inv} and crossing onset at $Q = 83$. Coloured points are the solved precipitation-onset states of the rate, compression, bromide and diffusivity families; they collapse onto the one onset line — the geometric content of the organizing coordinate, each within-electrolyte knob acting on S_{base} through the node it perturbs. Salting-out instead rescales c_{sat} and shifts the boundary itself (off this fixed- c_{sat} plane); electronic and electrolyte conductivity and flow do not move the operating point (grey dot). The solved field references the free monomer at $c_{\text{sat}}/3$ (the slope-3.00 normalization of Fig. 8b); the onset locus is plotted in that solved convention.

Free- I_2 transport (diffusivity prior)

Free- I_2 diffusivity is the single most leveraged knob in the model, and the reason is structural: it enters S *only* through the generation term, $\Pi_{\text{gen}} \propto 1/D_{\text{eff}} \propto 1/D_0$, leaving Π_{inv} and β untouched. Physically, Π_{gen} is the diffusion-polarization overshoot — the excess free- I_2 concentration that must accumulate at the pore wall to drive the electrogenerated flux $i_{\text{app}}/2F$ outward through the Nernst layer δ .

Halving the species diffusivity therefore doubles the required local pore-wall overshoot, so the surface concentration overtakes c_{sat} far sooner: precipitation onset advances from the baseline $Q_{S \geq 1} = 83.1$ to 45.1 (the *volume-averaged* peak stress is nearly unchanged, $S_{\text{avg,peak}} = 1.78$ vs. 1.76 baseline — it is the onset shift, not the averaged peak, that registers the doubled local overshoot, the same averaging caveat that applies to the rate and wetted-area knobs below), the solid inventory grows $52\times$ ($\epsilon_{s,\text{end}} : 3.22 \times 10^{-3} \rightarrow 1.67 \times 10^{-1}$), block-

ing saturates ($\theta_{\text{end}} \rightarrow 1.000$, $A_{\text{bare,end}} \approx 2 \times 10^{-8}$), pore permeability falls 74 % ($K_{\text{perm,end}} : 0.956 \rightarrow 0.252$), and the accessibility loss lifts the cell voltage ($V_{\text{end}} : 1.729 \rightarrow 1.849 \text{ V}$).

Doubling transport is the exact mirror through the same $1/D$ scaling: the overshoot halves ($S_{\text{avg,peak}} : 1.76 \rightarrow 1.22$), onset is pushed to $Q_{S \geq 1} = 106.2$, essentially no solid forms ($\epsilon_{s,\text{end}} = 7.3 \times 10^{-5}$, $44\times$ below baseline), and two-thirds of the area survives ($A_{\text{bare,end}} = 0.666$). The end-state solid inventory thus spans $\sim 2290\times$ across the $\pm 2\times$ diffusivity bracket. In the ranked sensitivity the diffusion prior is first of nine families (effect_{total} = 20.24; dominant node solid-inventory), roughly twice the next knob and five times current density. Because Π_{gen} is the largest mobile term in S at the operating current and scales as $1/D$, transport sets both the precipitation *onset* and the post-onset blocking *severity* simultaneously — making it the master lever on the dissolution-limited supersaturation pathway.

The principal caveats are that the scan resolves only the two $\pm 2\times$ points at fixed $Q = 120$, so the continuous $S(D)$ law is inferred rather than mapped; that D_{eff} folds in the $\epsilon_1^{1.5}$ factor so a porosity change would compound this sensitivity; and that the accompanying voltage swing is a *consequence* of accessibility loss, not an independent transport effect. The diffusivity prior is correspondingly the dominant source of parametric uncertainty in any absolute lifetime estimate, so the load-bearing conclusion is the ranking, not a calibrated D_0 .

Applied current density (rate)

Applied current density is the leading *operating* accelerant because it is the only operating variable that enters the generation group $\Pi_{\text{gen}} = i_{\text{app}}\delta/(2FD_{\text{eff}}c_{\text{sat}})$ directly and linearly, with δ , D_{eff} and c_{sat} all rate-independent, so $dS/di_{\text{app}} > 0$ at fixed β . In the throughput-normalized scan ($Q = 40$), the peak stress rises from $S_{\text{avg,peak}} = 0.328$ at 20 mA cm^{-2} to 0.483, 0.924 and 1.705 at 40, 80 and 120 mA cm^{-2} — a 5.2-fold stress amplification for a 6-fold rate increase.

The system is strictly subcritical ($S < 1$, no precipitation) through 80 mA cm^{-2} and crosses the onset $S = 1$ only between 80 and 120 mA cm^{-2} at this fixed 40 mA h cm^{-2} throughput. This model onset-within-a-fixed-capacity is a distinct quantity from the *steady* dissolution-limited current ceiling $i^* = 2Fk = 232\text{--}425 \text{ mA cm}^{-2}$ obtained from the independent kinetics of Zhao *et al.* (§2.6); the two should not be equated. The crossing signature is abrupt: solid inventory jumps more than three orders of magnitude ($\epsilon_{s,\text{end}} : 2.5 \times 10^{-7} \rightarrow 4.8 \times 10^{-4}$, $\sim 1900\times$, between 80 and 120 mA cm^{-2}), accessible area collapses from $A_{\text{bare,end}} = 0.999$ to 0.220 (i.e. $\theta_{\text{end}} = 0.78$, 78 % blocking), and the voltage runs to $V_{\text{end}} = 1.854 \text{ V}$ with the working-window span widening

14.7× (0.017 V to 0.254 V). The threshold capacities materialize only at 120 mA cm⁻² ($Q_{S=1} = 8.7$, $Q_{\theta=0.5} = 31.3$), confirming a sharp onset.

Physically, current density fixes the local electro-oxidation flux of I⁻ to free I₂: below the clearance ceiling set by diffusion plus complexation the free-I₂ concentration stays under c_{sat} and no solid forms, but once generation outruns dissolution-limited resupply the liquid supersaturates, I₂(s) precipitates onto the carbon, and the surviving sites are forced to higher overpotential. The failure is sharper than the linear Π_{gen} term implies because high rate simultaneously depletes the complexation buffer — β_{end} falls from 1132 to 565 (2.0×) across the sweep — shrinking the denominator of S as the numerator grows.

In the ranked analysis current density carries effect_{total} = 4.10 with a stress-onset-dominated profile, making it the most influential lever among the directly comparable *true-solve-clean* families (compression, the next clean family, scores 1.01). Caveats: the static- β theory under-predicts the true high-rate steepening; the high-rate branch (160 mA cm⁻² to 320 mA cm⁻²) failed to converge so the $S = 1$ crossing is bracketed rather than pinned; part of the V_{end} rise is ordinary ohmic/activation overpotential, so accessibility and solid inventory — not voltage — are the failure-defining outputs; and the two inherited-risk families above (diffusion prior, salting-out) score higher raw totals on a different provenance footing, so current density is the leading *operating-condition* knob within the clean set rather than the unconditional global maximum.

Mechanical compression

Compression enters the theory through two distinct, opposing terms while leaving the chemical thresholds untouched. Compressing the felt simultaneously thins it ($L : 3.0 \rightarrow 1.5$ mm) and lowers the porosity ($\varepsilon_l : 0.900 \rightarrow 0.800$). Through the Bruggeman relation $D_{\text{eff}} = D_0 \varepsilon_l^{1.5}$ a lower porosity throttles intra-pore transport and so raises the generation stress $\Pi_{\text{gen}} \propto 1/\varepsilon_l^{1.5}$ (through D_{eff} only; L enters not here but the separate pore-volume capacity below); this transport channel is, however, weak and not cleanly monotonic, because the ^{1.5} exponent over the realistic porosity window changes D_{eff} by only 0.93 to 1.11× while L co-varies in the same direction. The simulated peak stress is consequently flat: $S_{\text{avg,peak}}$ spans only 1.7214 ($L=3.0$) to 1.7590 (baseline $L=2.0$), with the tightest felt ($L=1.5$) sitting marginally *below* baseline at 1.7570 — a <2% non-monotonic excursion.

The decisive channel is geometric: compression shrinks the accessible pore volume εL (computed from the solve's own (L, ε) pairs, εL relative to the $L=2.0$ baseline is 0.71/1.29/1.59× for $L=1.5/2.5/3.0$ mm), the reservoir the

precipitate must fill before it occludes the pore. Packing the same precipitated mass into a smaller volume drives the occluded fraction toward unity sooner: at $Q = 120$ the tight felt reaches $\theta_{\text{end}} = 0.9998$ with negligible retained area ($A_{\text{bare,end}} = 1.95 \times 10^{-4}$), whereas the loose felt holds $\theta_{\text{end}} = 0.9204$ and $A_{\text{bare,end}} = 7.93 \times 10^{-2}$ — a ~407× difference in surviving accessibility, and unlike the stress channel this trend is strictly monotonic. Equivalently, the blocking onset rises monotonically $Q_{\theta=0.5} : 97.6 \rightarrow 103.3$ mA h cm⁻² (+5.9%) as the felt is relaxed, while the precipitation onset shifts only marginally ($Q_{S=1} : 82.4 \rightarrow 84.3$).

The flat onset and the large deep-charge spread are the same reservoir law read at different depths of charge, and resolve an apparent tension with $Q_f \propto \varepsilon L$. The idealized reduction behind that scaling (Eq. 13 with $\varepsilon_s \propto Q_C/L$ and threshold $\varepsilon_s^* \propto \varepsilon$) assumes clean linear accumulation and would make even the half-blocking onset scale with the pore volume; in the full solve, however, θ_{eff} reaches 0.5 while ε_s is still tiny ($\sim 1 \times 10^{-4}$, total basis, varying <9% across the series), so the onset is fixed before the compression branches diverge and stays nearly geometry-independent (+5.9%). The pore-volume leverage instead emerges at deep charge—the ~400× spread in surviving accessibility above—and in the experimental usable capacity (Fig. 9a), the regime to which the εL law actually refers; the $(S-1)_+$ variation and dissolution back-reaction omitted from the idealized reduction are what flatten the onset while leaving the deep-fill reservoir scaling intact. The native closure therefore under-resolves this leverage at the half-blocking onset, which is why the quantitative εL law is anchored to the theory's deep-fill limit and the experiment rather than to the native onset capacity.

The solid-inventory (capacity) node dominates the ranked sensitivity (weight 0.623, 62 % of effect_{total} = 1.007), confirming that looser compression protects chiefly by enlarging the precipitate reservoir, with a secondary transport assist from higher D_{eff} . Crucially, compression leaves Π_{inv} and β nearly untouched (β_{end} : 553.9 to 561.5, <1.4%), so it modulates only how fast and how completely the pore is driven into the blocking regime, never the chemistry of onset. The non-monotonic V_{end} (1.670 to 1.776V, peaking at $L=2.5$) is an ohmic/path-length artifact and carries no failure information; the internal-overpotential maps (Fig. 12) show the electronic/contact term growing toward the most-compressed felt. The overall leverage is small (effect_{total} = 1.007), so compression is best framed as a fine geometric trade-off rather than a primary mitigation lever; the $L=2.0$ point is the baseline solve (not a compression-branch solve), and because L and ε co-vary along the physical compression locus the transport ($\varepsilon^{1.5}$) and capacity (εL) contributions are inferred from the group definitions rather than orthogonally resolved. Com-

pression is therefore not used here as a one-parameter proof of pore-volume control; its mechanistic value is that it separates iodine retention, blocking efficiency and voltage/contact compensation in a real electrode geometry, reinforcing that retained iodine is not blocking iodine (Discussion).

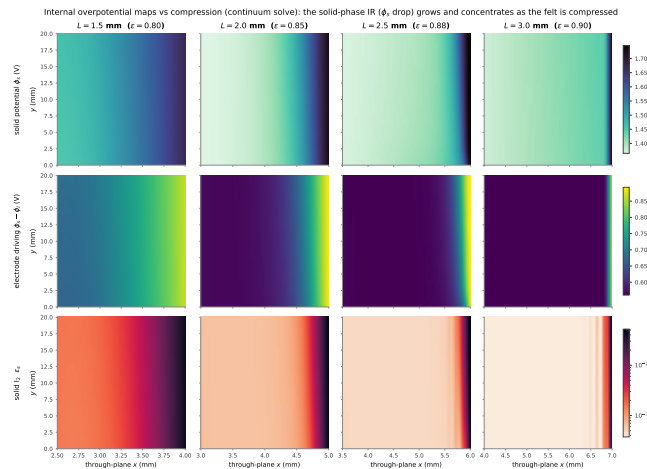


Figure 12 Internal overpotential maps vs compression (native continuum solve): rows show the solid-phase potential (ϕ_s) drop, the electrode driving potential ($\phi_s - \phi_\ell$) and the solid I_2 fraction (ε_s) across $L = 1.5$ – 3.0 mm; the electronic/contact penalty grows and concentrates toward the collector as the felt is compressed.

Active-material inventory

Charged iodine inventory (state of charge) is the state variable whose accumulation drives the onset, because it sets the absolute quantity of free I_2 that the finite-volume, saturation-limited pore liquid must accommodate, entering S through the inventory group $\Pi_{\text{inv}} = c_{I_2, \text{tot}}/c_{\text{sat}}$. With c_{sat} fixed and Π_{gen} held constant across the sweep, charging raises $c_{I_2, \text{tot}}$ so that $\Pi_{\text{inv}} \sim Q/Q_{\text{base}}$ (a state-of-charge ramp on the baseline configuration, not an independent iodide molarity variation — see the caveat below). The native capacity-window scan makes this quantitative and reveals a sharply super-linear, threshold-type response: the precipitated solid fraction climbs from $\varepsilon_{s, \text{end}} = 9.0 \times 10^{-8}$ at $Q = 80 \text{ mA h cm}^{-2}$ to 1.25×10^{-4} at 100 and 3.22×10^{-3} at 120 mA h cm^{-2} — a $25.8\times$ jump in solid for only a 20 % inventory increase between 100 and 120 mA h cm^{-2} , and $3.6 \times 10^4\times$ across the full window — because near $S = 1$ a modest inventory increment converts a large excess of supersaturated free I_2 into $I_2(s)$.

The dimensionless theory localizes the two operative thresholds: precipitation onset at $Q_{S=1} = 83.1 \text{ mA h cm}^{-2}$ (independently recovered as the baseline $Q_{S \geq 1} = 83.11$) and functional blocking at $Q_{\theta=0.5} = 99.67 \text{ mA h cm}^{-2}$, in quantitative agreement with the simulated accessibility, whose blocked fraction crosses $\theta_{\text{eff}} = 0.5$ right at 100 mA h cm^{-2} ($\theta_{\text{end}} : 0.335 \rightarrow 0.519 \rightarrow 0.824$ over $Q = 96 \rightarrow 100 \rightarrow 108$)

and saturates at 0.990 by 120 mA h cm^{-2} . The accessibility collapse, not any conductivity change, drives the end-of-life overpotential ($V_{\text{end}} : 1.427 \rightarrow 1.454 \rightarrow 1.601 \rightarrow 1.729 \text{ V}$ over $Q = 80 \rightarrow 100 \rightarrow 115 \rightarrow 120$).

That the solid-inventory node dominates failure is corroborated across the ensemble: the two largest-effect families both report solid-inventory as their dominant node (node weights 16.68 and 6.49), but they reach it through *different* terms of S — the diffusion prior (20.24) through D_{eff} in Π_{gen} , and the salting-out activity (9.55) through c_{sat} in Π_{inv} — confirming that whether one throttles removal or shrinks the solubility headroom, the failure is registered as solid accumulation. Each knob that moves the operating point rescales failure by changing the free- I_2 supply/clearance balance or the pore volume available to hold the precipitate, so moderating the usable inventory and raising c_{sat} are the most direct routes to keeping $S < 1$.

Two caveats temper the magnitude: Q is a charge-throughput/SOC proxy swept on the baseline configuration rather than an independent molarity variation (with the $S = 1$ and $\theta = 0.5$ thresholds imported from the baseline-config theory rows), and the static β does not track the I^- consumed on charge, so the model conservatively under-predicts the denominator erosion and the true inventory sensitivity is likely even steeper than the simulated $25.8\times$ solid jump.

Salting-out (apparent solubility)

The saturation concentration c_{sat} is not a fixed material constant but an *apparent* solubility set by the supporting electrolyte: the concentrated $\text{ZnI}_2/\text{NH}_4\text{Br}$ medium salts I_2 out of solution, lowering c_{sat} and shrinking the solubility headroom. Because c_{sat} enters S both as the denominator of the inventory group $\Pi_{\text{inv}} = c_{I_2, \text{tot}}/c_{\text{sat}}$ and (through $D_{\text{eff}}c_{\text{sat}}$) of the generation group, a lower c_{sat} raises the stress on both channels and brings $S = 1$ forward. In the cross-knob ensemble this makes salting-out the second-ranked family overall (effect 9.55, behind only the free- I_2 diffusivity prior at 20.24), and like the inventory knob it registers its impact predominantly at the solid-inventory node (node weight 6.49). The lever is therefore two-sided: additives that salt I_2 *in* — cosolvents or complexing agents that raise the apparent c_{sat} — are a direct route to keeping $S < 1$, the chemical counterpart to raising the speciation buffer β . We treat c_{sat} as a bounded constant here (1.33 mol m^{-3}); it is the most consequential untested parameter in the theory (Discussion, bound iv).

Supporting bromide

Supporting bromide enters the theory exclusively through the complexation buffer in the denominator of S , $\beta = 1 + K_{\text{ICl}^-} +$

$K_{\text{Br}}c_{\text{Br}^-}$, so that $\partial S/\partial c_{\text{Br}} = -(\Pi_{\text{inv}} + \Pi_{\text{gen}})K_{\text{Br}}/\beta^2 < 0$. Crucially NH_4Br perturbs neither numerator driver — not Π_{gen} and not Π_{inv} — so it acts purely as a speciation buffer, complexing dissolved I_2/I_3^- and dividing down the free- I_2 supersaturation. In the native scan (2 M, 3 M, 5 M) the buffer rises only modestly, $\beta_{\text{end}} = 543.2 \rightarrow 561.5$ (+3.4%), and the stress response is correspondingly weak: peak stress falls from $S_{\text{avg,peak}} = 1.772$ to 1.743 (~1.6%), shifting precipitation onset $Q_{S=1} : 81.2 \rightarrow 84.0$ and blocking $Q_{\theta=0.5} : 97.6 \rightarrow 100.7$ — gains of only ~3 capacity units (~3%).

The one appreciable signature is the precipitated inventory, which drops $\sim 3.9\times$ ($\varepsilon_{s,\text{end}} : 7.47 \times 10^{-3} \rightarrow 1.90 \times 10^{-3}$) with a small recovery of accessible carbon ($A_{\text{bare,end}} : 5.0 \times 10^{-3} \rightarrow 1.5 \times 10^{-2}$); accordingly the ranking assigns bromide a dominant node of solid-inventory but an $\text{effect}_{\text{total}}$ of just 0.71 — sixth of nine families and roughly $28\times$ weaker than the diffusion prior. The physical reading is instructive: because failure is governed by dissolution-limited free- I_2 generation against a fixed solubility ceiling, a buffer already near $\beta \approx 550$ has almost no marginal leverage — bromide can shave the supersaturation peak and thin the $\text{I}_2(\text{s})$ film, but it cannot remove the I_2 being produced, so θ_{eff} remains pinned near unity once onset is passed.

This modeled weakness carries an explicit caveat: the static, lumped β closure (constant K_{Br} , fixed c_{sat} and dissolution constant k) cannot represent concentration-dependent polybromide/polyiodide equilibria or the genuine rise in effective I_2 solubility in bromide-rich electrolytes, so the simulation almost certainly under-predicts the true protective magnitude of NH_4Br . The model is therefore reliable for the *direction* and *mechanism* (β -buffering of S , reduced ε_s) [38–40] but not for the quantitative size of the bromide benefit.

Electrode specific (wetted) area

The electrode specific (wetted) area a_s enters S heuristically through the generation group, since the locally relevant flux is the area-specific current density $i_{\text{local}} = i_{\text{app}}/(a_s L)$ (with a_s the volumetric area, m^{-1}), so that more wetted area dilutes the local generation and, to first order, $\Pi_{\text{gen}} \propto 1/a_s$ while Π_{inv} , β and the pore-volume capacity εL are left untouched; this is a local-flux interpretation, as a_s does not appear in the planar-Nernst-layer Π_{gen} of Eq. (11). We flag this knob as *mechanistically unresolved in the present scan*, because only the low-area branch converged. The single converged point ($a_s \times 0.5$) does shift the volume-averaged peak stress in the predicted direction ($S_{\text{avg,peak}} : 1.759 \rightarrow 1.801$, +2.4%) while leaving the precipitation onset pinned at $Q_{S \geq 1} = 83.1$ (onset is set by cumulative throughput, not by how current is distributed). Its end-state, however, runs *counter* to the naive “less area fails harder” expectation: at $Q = 120$ the

low-area cell shows *less* blocking and *more* surviving area ($\theta_{\text{eff}} : 0.990 \rightarrow 0.943$, $A_{\text{bare,end}} : 0.0104 \rightarrow 0.0566$), *less* deposited solid ($\varepsilon_{s,\text{end}} : 3.22 \times 10^{-3} \rightarrow 5.1 \times 10^{-4}$) and a *lower* voltage (V_{end} down 0.097 V). A consistent reading of this single point (lower exchange current $i_0 a_s$ spreading the same total charge over fewer but less-occluded sites) cannot be separated from a genuine reversal without the missing high-area branch, so we report a_s only as ranked mid-tier ($\text{effect}_{\text{total}} = 0.54$, dominant node solid-inventory) and defer its mechanistic sign to a dedicated converged a_s sweep.

Non-drivers: conductivity and flow (a mechanistic contrast)

Liquid conductivity (σ_l), carbon-felt electronic conductivity (σ_s), and electrolyte flow rate (Q_{flow}) form a distinct class of *voltage-channel* parameters that, by construction, cannot appear in S . The conductivities enter only the charge-transport law ($i = -\sigma_l \nabla \phi_l$, $i = -\sigma_s \nabla \phi_s$) and therefore set the ohmic overpotential and cell voltage, while Q_{flow} touches only the external boundary layer; none alters the per-coulomb I_2 generation, c_{sat} , δ , D_{eff} , or $\varepsilon_l L$. The scan demonstrates this decoupling quantitatively at fixed galvanostatic $Q = 120$: varying σ_l over 10 S m^{-1} to 40 S m^{-1} swings V_{end} by 77 mV yet leaves the onset unchanged at $Q_{S \geq 1} = 83.1$ (spread 0.00, $\Delta \theta_{\text{end}} = 3 \times 10^{-4}$); σ_s over 80 S m^{-1} to 320 S m^{-1} produces the largest ohmic excursion in the whole study (141 mV) but again pins the onset at 83.1; and Q_{flow} (25 mL min^{-1} to 100 mL min^{-1}) moves V_{end} by only 8 mV and $Q_{S \geq 1}$ by 0.22.

The ranking places these at the bottom of the spectrum — σ_l is the sole parameter whose dominant node is *voltage* ($\text{effect}_{\text{total}} = 0.12$), with Q_{flow} (0.20) and σ_s (0.84) close behind — one to two orders of magnitude below the true drivers. The single instructive nuance is σ_s : a poorly conducting felt redistributes the reaction toward the membrane side, locally raising the I_2 flux and thus the end-state precipitated inventory ($\varepsilon_{s,\text{end}} = 7.2 \times 10^{-3}$ at 80 S m^{-1} vs. 1.6×10^{-3} at 320 S m^{-1}), but this is post-onset current crowding rather than a shift of the supersaturation threshold, which is precisely why $Q_{S \geq 1}$ does not move.

The mechanistic conclusion is itself a result: iodine passivation lives in the stress channel (S) whereas these parameters act on the orthogonal voltage channel (Ohm’s law), so ohmic engineering buys round-trip efficiency but *cannot* cure the dissolution-limited supersaturation failure — a clean negative control that sharpens identification of the genuine drivers. (Viscosity, conceptually a member of this class via Q_{flow}/D , is omitted from the numerical claim because its scan rows failed to converge.)

Design corollary

Read through Eq. (11), the knob ensemble yields a single, self-consistent design rule. Because failure is the crossing of $S = 1$ followed by the growth of θ_{eff} to blocking, mitigation must (i) lower the generation numerator Π_{gen} — improve free- I_2 transport (D_{eff} , by far the dominant lever), reduce the local flux i_{app}/a_s via moderate rate and high wetted area, and avoid over-compressing the felt; (ii) raise iodine accommodation $\Lambda_{\text{Br}}^{\text{rel}}$ with supporting bromide (its dominant cross-composition effect, a $\sim 2.5\times$ gain to the intrinsic ceiling) while maintaining the within-cycle buffer β_{intra} ; and, most directly, (iii) widen the gap between the precipitation onset $Q_{S=1}$ and the functional-blocking capacity $Q_{\theta=0.5}$ — i.e. retain accessibility even after the first solid forms — by enlarging the precipitate reservoir $\varepsilon_l L$ and limiting the usable iodide inventory that feeds Π_{inv} . Conductivity and flow optimize voltage and efficiency but lie outside this stress channel and cannot substitute for it. This priority order is consistent with recent ZIFB bottleneck and high-current electrode-material discussions, which frame electrolyte/intermediate control and electrode utilization as central engineering limits [8,37].

External consistency

The model trends above were checked against our full-cell measurements, which are reported in full in a separate experimental study; a representative operating-regime subset is tabulated in Supplementary Table S3, and here these data serve only as an external-consistency anchor, not as a validation of the model. Any short potential-step (CA/EIS) perturbation is likewise treated only as a low-dose, short-time accessibility check; it is not used to claim that early iodine is intrinsically non-passivating, and it does not rule out passivation after larger retained-iodine inventory, longer exposure, or morphology/connectivity changes — the regime the continuum model assigns to the high-capacity blocking threshold.

Experimental chemistry axis and the bromide-accommodation correction

The composition knobs are the one place where the lumped buffer β of Eq. (1) proves insufficient, and full-cell electrolyte sweeps make the correction concrete (Fig. 13). At fixed $1 \text{ mol L}^{-1} \text{ ZnI}_2$ and matched current, raising supporting bromide from 0 to 4 mol L^{-1} lifts the attainable areal capacity $\sim 2.5\times$ ($54.6 \rightarrow 137.9 \text{ mAh cm}^{-2}$), saturating by $2\text{--}3 \text{ mol L}^{-1}$; raising ZnI_2 at fixed $4 \text{ mol L}^{-1} \text{ NH}_4\text{Br}$ instead lowers it monotonically ($98.5 \rightarrow 57.1 \rightarrow 45.1 \text{ mAh cm}^{-2}$). The buffer β cannot reproduce either trend: with K_{ICl^-} dominant, β shifts only $\sim 3\%$ across the bromide sweep (and its c_1 dependence largely cancels against Π_{inv} on the iodide axis), so a β -only stress

predicts a near-flat capacity (Fig. 13c, $0.97\times$ vs. the observed $2.5\times$).

The failure capacity follows the composition-resolved stress S_{eff} of Eq. (11) through the capacity-ceiling closure

$$Q_f = \min(Q_{\text{cap}}, A/(S_{\text{eff}} - 1)_+). \quad (19)$$

Calibrating the composition dependence directly, with the rate kept explicit, gives the absolute stress

$$S_{\text{eff}} = \frac{\kappa i_{\text{app}} (1 + \chi_I^{\text{app}} c_{\text{I}^-})}{c_{\text{sat}} (1 + K_{\text{sol,Br}}^{\text{app}} c_{\text{Br}^-})}, \quad (20)$$

whose bromide and iodide factors are exactly the *apparent iodine-accommodation* term $\Lambda_{\text{Br}}^{\text{rel}} \propto (1 + K_{\text{sol,Br}}^{\text{app}} c_{\text{Br}^-})$ (bromide raising the effective I_2 solubility) and the excess-iodide index $\chi_{\text{ZnI}_2}^{\text{rel}} \propto (1 + \chi_I^{\text{app}} c_{\text{I}^-})$ of Eqs. (11)–(12); the prefactor κ absorbs the baseline constant $S_{\text{base}}(c_0)$, so the absolute and baseline-referred forms differ only by that constant. One parameter set fits both axes ($R^2 = 0.998$) and far outperforms the β -only null ($R^2 = 0.05$). Crucially, $K_{\text{sol,Br}}^{\text{app}} \approx 0.5 \text{ M}^{-1}$ is *not* the microscopic $\text{I}_2 + \text{Br}^- \rightleftharpoons \text{I}_2\text{Br}^-$ association constant ($\sim 10 \text{ M}^{-1}$): it is a capacity-level coefficient lumping solubility, I_2Br^- -mediated dissolution, activity and deposit morphology/reversibility, which these data cannot separate. We stress that $\Lambda_{\text{Br}}^{\text{rel}}$ is an effective accommodation factor for the low-to-moderate-bromide, two-electron I^-/I_2 regime modeled here; in strongly halide-mediated cells that unlock a higher-valent I_2/I^+ couple, part of the bromide capacity gain is a distinct faradaic step and a pathway switch (interhalogen I^+ stabilization) rather than raised I_2 solubility [10,41,42], so $\Lambda_{\text{Br}}^{\text{rel}}$ should not be extrapolated to high-bromide four-electron operation. A further, structural confound is that the composition axis varies NH_4Br as a whole salt, which is *trifunctional* in this chemistry: Br^- complexes I_2 to soluble I_2Br^- (the positive-electrode iodine accommodation modelled here), NH_4^+ complexes Zn^{2+} and suppresses zinc-dendrite growth on the *negative* electrode, and the added salt raises the electrolyte conductivity [43]. Because Br^- is never decoupled from NH_4^+ in our cells (only ZnI_2 and NH_4Br were used), the measured $\sim 2.5\times$ gain is a *cell-level* response that these data cannot partition between positive-electrode accommodation, negative-electrode zinc improvement and reduced ohmic loss; $\Lambda_{\text{Br}}^{\text{rel}}$ is accordingly an apparent capacity-level factor, not a positive-electrode solubility per se. The positive-electrode precipitation picture does not rest on this axis: it is carried by the NH_4Br -invariant iodide axis (where NH_4^+ and conductivity are held fixed while only the positive-side iodine inventory varies) and by the dissolution-limited physics anchored to independent kinetics. A bootstrap/leave-one-out identifiability test (Fig. 13e) sets what may be claimed: the intrinsic ceiling $Q_{\text{cap}} \approx 137 \text{ mAh cm}^{-2}$ and $K_{\text{sol,Br}}^{\text{app}}$ are robustly identified and stable across capacity definitions, whereas A, κ

trade off and the iodide coefficient χ_I^{app} is only weakly constrained — the iodide axis fits three active parameters to three points and is confounded by the large ionic-strength change from 1 to 3 mol L⁻¹ ZnI₂. We therefore report the bromide-accommodation leg quantitatively and the iodide leg as a directional trend, not an identified constant. The core picture is unchanged: free-I₂ stress → I₂(s) → accessibility loss; only the cross-composition chemical entry is reassigned from β to Λ_{Br} and χ_I^{app} .

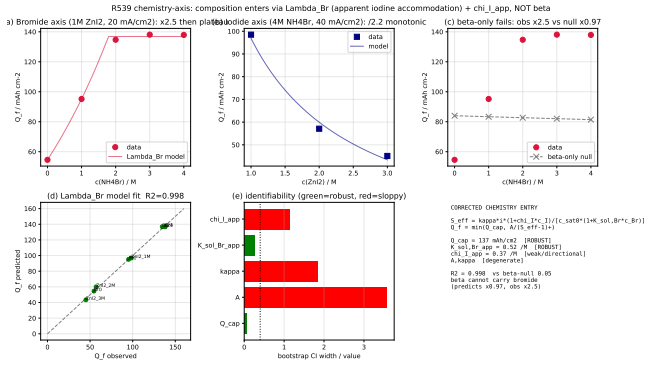


Figure 13 Experimental chemistry axis and the bromide-accommodation correction. (a) Areal capacity vs. supporting bromide (1 M ZnI₂, 20 mA cm⁻²): $\sim 2.5\times$ rise saturating by 2–3 M. (b) vs. ZnI₂ (4 M NH₄Br, 40 mA cm⁻²): monotonic $\div 2.2$ decline. (c) the β -only stress predicts a near-flat bromide response ($0.97\times$) against the observed $2.5\times$. (d) the Λ_{Br} model fits both axes, $R^2 = 0.998$. (e) bootstrap identifiability: Q_{cap} and $\chi_{\text{sol,Br}}^{\text{app}}$ robust; Λ_{Br} , χ_I^{app} sloppy. Capacities are max-deliverable from raw .ndax; full cells 4 cm², three-electrode tests (Ag/AgCl, positive felt) 1 cm².

6 Discussion

The theory reorganizes a large body of device phenomenology around one dimensionless stress and two thresholds. Three consequences follow. First, “retained $\neq$ blocking” is no longer a slogan but a measurable gap ΔQ between $S = 1$ and ε_s^* ; the design objective is to widen it. The same supersaturation–precipitation–accessibility-loss logic recurs in other aqueous battery electrodes — local supersaturation precipitates a solid that occludes the active surface, as established for ZnO films on zinc anodes and for solubility-controlled Li₂S passivation of sulfur cathodes [44, 45] — so the framework is not iodine-specific. Made quantitative (Fig. 14), this universality is itself a design principle: the inventory stress $\Pi_{\text{inv}} = c_{\text{I}_2, \text{tot}}/c_{\text{sat}}$ scales inversely with the solubility headroom c_{sat} of the condensable intermediate, so iodine is structurally the most exposed electrode. The aqueous solubility of I₂ ($c_{\text{sat}} = 1.33 \text{ mol m}^{-3}$ [24]) is $\sim 14\times$ below that of dissolved sulfur in Li–S ($\approx 18 \text{ mol m}^{-3}$ [46]) and $\sim 320\times$ below that of Br₂ (430 mol m^{-3} , i.e. 0.43 mol L^{-1} [47]), so at equal oxidized inventory the iodine electrode bears that many times the super-saturation stress — which is

why ZnI₂ chemistry is precipitation/passivation-limited while the far more soluble ZnBr₂ fails instead by bromine crossover and shuttle [48]. The framework also names the chemical relief: the large I₃⁻ complexation buffer ($K \approx 698 \text{ M}^{-1}$ [12]) inflates the soluble inventory through β — precisely the lever Eq. (11) isolates. Complexation engineering, such as quaternary-ammonium additives that raise this accommodation, is one practical route to that relief [49].

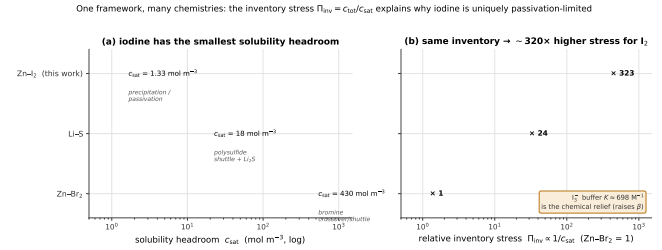


Figure 14 Cross-chemistry universality of dissolution-limited passivation. (a) Solubility headroom c_{sat} of the condensable intermediate for three conversion chemistries (aqueous for I₂/Br₂; an illustrative non-aqueous dissolved-sulfur model cap for Li–S): I₂/ZnI₂ (1.33 mol m^{-3} [24]), Li–S dissolved sulfur ($\approx 18 \text{ mol m}^{-3}$ [46]) and Br₂/ZnBr₂ (430 mol m^{-3} [47]). (b) The implied inventory stress $\Pi_{\text{inv}} \propto 1/c_{\text{sat}}$ at fixed oxidized inventory (normalized to ZnBr₂): iodine bears $\sim 320\times$ the stress of bromine and $\sim 14\times$ that of Li–S, the structural reason ZnI₂ is uniquely precipitation/passivation-limited. The I₃⁻ complexation buffer ($K \approx 698 \text{ M}^{-1}$ [12]) is the chemical relief identified by the buffer β . Values are external literature data, not model outputs.

Second, the tested levers act on *different nodes* of the stress framework rather than a single term: current density and compression enter the generation stress Π_{gen} (the latter through D_{eff}), state of charge and iodide inventory enter Π_{inv} , within-cycle speciation enters β_{intra} , and electrolyte composition enters the cross-composition factors $\Lambda_{\text{Br}}^{\text{rel}}$ (supporting bromide, cosolvents that solvate I₂ [16, 38–40]) and $\chi_{\text{ZnI}_2}^{\text{rel}}$. Lowering Π_{gen} and raising iodine accommodation both push the precipitation onset to higher capacity, whereas increasing the iodide inventory without accommodation only raises Π_{inv} and fails earlier. Third, the dissolution-limited current $i^* = 2Fk$ is the natural rate ceiling, and engineering k (chemistry) is more effective than engineering the retained amount (geometry). This study isolates the positive-electrode *precipitation–passivation* channel; it is complementary to the iodine crossover/shuttle and membrane losses addressed elsewhere [21, 40], which act on inter-electrode transport rather than on the surface super-saturation stress S . The picture is bounded, and we state the bounds precisely. (i) The continuum model is a mechanism scaffold, not an exact voltage validator (absolute V_{end} is over-blocked by $\sim 0.24 \text{ V}$). (ii) The two-threshold capacities are computed under the empirical accessibility closure; swapping it for the physical mesoscale closure preserves the ordering but changes

the absolute blocking magnitude, so Q_f and the ΔQ gap are closure-conditional. (iii) The functional-blocking solid fraction ε_s^* is calibrated, and its fiber-model cross-check shares the same accessibility mapping, so the agreement is consistency, not full independence. (iv) The saturation c_{sat} is fixed at 1.33 mol m^{-3} ; it sets $S = 1$ linearly and is the most consequential untested constant. (v) The static buffer β captures the direction but not the magnitude of the chemistry lever; quantitative magnitudes come from the dissolution kinetics. (vi) The pore-volume capacity scaling rests on $n = 4$ points ($p = 0.06$). (vii) We analyse the accessibility-collapse failure; the deep-post-failure hydraulic runaway — full pore clogging, diverging pressure, and the mechanical tubing/seal rupture seen in practice — is a distinct downstream mode outside this scope. Within these bounds the theory is internally consistent, mechanistically coherent, and externally anchored to the independent dissolution kinetics of Zhao *et al.* [16].

7 Conclusion

We have shown that positive-electrode passivation in zinc-iodine flow batteries is a dissolution-limited super-saturation phenomenon governed by a single dimensionless stress S_{eff} (baseline $S_{\text{base}} = (\Pi_{\text{inv}} + \Pi_{\text{gen}})/\beta_{\text{intra}}$) and a two-threshold (precipitation, functional blocking) criterion. The theory reproduces the model's exported super-saturation field as an internal-consistency check, organizes the tested operating and design variables, recovers the failure-capacity and rate-threshold scalings, and is anchored to independently measured dissolution kinetics. Crucially, the pre-failure voltage rise is dominated by iodide-depletion concentration overpotential, with accessibility loss entering only late, so the absolute voltage cannot by itself demand a coalesced passivating film—the blocking morphology must instead be inferred from the mesoscale geometry and percolation. It thus separates iodine that is present from iodine that blocks from iodine that exacts a voltage penalty, and it prescribes the design map—suppress generation stress, raise iodine accommodation (keep the within-cycle buffer), do not chase iodine amount. The same retained-versus-blocking accounting should extend to the emerging high-valence, multielectron halogen-conversion chemistries [42], where still more oxidized halogen must be stored without occluding the electrode.

Methods

Density-functional theory. Periodic DFT (CP2K/Quickstep, PBE–D3(BJ), DZVP-MOLOPT-SR-GTH basis with GTH

pseudopotentials, plane-wave cutoff 150 Ry/relative cutoff 30 Ry, Γ point) on graphitic slabs and ribbons with oxygenated defects; geometries optimized, single-point energies counterpoise-corrected for basis-set superposition error, and open-shell sites (single vacancy) treated spin-polarized (LSD, Fermi smearing, 80 added states). Single- I_2 adsorption energies referenced to gas-phase I_2 (−0.40 basal, −0.85 carbonyl, −1.00 hydroxyl, −0.32 vacancy eV) and an optional 2I_2 coalescence (−0.69 eV at C–OH) are used as bounded mechanism priors; finite-cluster water-pair energies are retained as diagnostics only and excluded as unreliable. **Molecular dynamics.** Classical force-field MD (GROMACS) of $1 \text{ M ZnI}_2 + 4 \text{ M NH}_4\text{Br}$ in SPC/E water with explicit oxidized carriers (neutral two-site I_2 ; linear $\text{I}_3^-/\text{I}_2\text{Br}^-$), using Joung–Cheatham halide, 12–6 Zn^{2+} and GAFF/OPLS-type NH_4^+ parameters; Packmol-built boxes, energy minimization, NVT/NPT equilibration and long production; diffusivities from mean-squared displacement (Einstein relation), with electronic-continuum charge scaling ($q \approx 0.8$) bracketed by formal charges ($q = 1.0$); reported as a bounded prior. **Mesoscale closures.** Cylindrical fiber unit cell ($r_f = 7.5 \mu\text{m}$, $\varepsilon = 0.85$) with cylindrical finite-difference diffusion ($8 \times 36 \times 36$) for $\theta_{\text{eff}}(\varepsilon_s)$; 15^3 pore-throat network (log-normal radii, sparse SPD solve) for $K(\varepsilon_s)$. **Continuum model.** Finite-element tertiary current distribution with native two-way Darcy/porous-media flow (velocity solved and coupled into transport) (COMSOL Multiphysics 6.3, implicit BDF time stepping); galvanostatic $i_{\text{app}} = 400 \text{ A m}^{-2}$; speciation K_1, K_{Br} ; Butler–Volmer kinetics with a precipitation/dissolution source and algebraic coverage/accessibility feedback closures; the full production constant set and solved cases accompany the deposited archive. **Reduced model.** Eq. (18) fitted by linear least squares in its kinetic parameters to the real pre-failure charge curve. **Experimental anchor.** Capacity-limit cycling of 4 cm^2 cells at $\sim 40 \text{ mA cm}^{-2}$, cycled to clean cycles; used only to set the operating regime, the voltage scale and the knob trends. All scripts and frozen source data accompany the manuscript.

Data and code availability

All analysis and figure scripts, the continuum-model field exports (CSV), the reduced-model fits, the parameter-scan tables and the per-figure source data are provided as a versioned archive with a QA manifest (checksums and provenance for each figure); the repository DOI will be provided on acceptance. The raw solved continuum (.mph) cases are large; representative cases are deposited per journal policy and the full set is available from the authors on reasonable request.

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Conflict of interest The authors declare that they have no conflict of interest.

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